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Review

An insight into allele-selective approaches to lowering mutant huntingtin protein for Huntington's disease treatment

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ABSTRACT

Huntington's disease (HD), a monogenic neurodegenerative disorder, stems from a CAG repeat expansion within the mutant huntingtin gene (*HTT*). This leads to a detrimental gain-of-function of the mutated huntingtin protein (mHTT). As of now, there exist no efficacious therapies to alter the disease progression. In view of the monogenetic mutation nature and an indispensable role of wild-type *HTT* in healthy neurodevelopment and cellular functions, the developing strategy of allele-selectively deleting/silencing mutant *HTT* as well as only inactivating mHTT without altering wild-type *HTT* or wild-type huntingtin protein (wtHTT) comes highly recommended, and may offer a promising treatment option for HD. Here, we reviewed the therapeutic approaches that allele-selective lowering mHTT expression by targeting only mutant *HTT* DNA, RNA and mHTT along with recent preclinical and clinical outcomes and challenges, in anticipation of some novel ideas to be introduced into HD therapeutic research.

1. Introduction

Huntington's disease (HD) is an autosomal neurodegenerative disorder characterized by a dominant inheritance pattern, giving rise to distinctive clinical symptoms such as involuntary movements, cognitive deterioration, and psychiatric disturbances. Despite numerous research efforts on HD treatment, there are currently no FDA-approved disease-modifying treatment to target the disease causes [1,2]. HD is caused by an aberrant expansion of CAG repeats in the exon 1 of huntingtin gene (*HTT*) encoding a polyglutamine (polyQ) stretch at the *N*-terminus of huntingtin protein (HTT). Most of HD patients carry a wild-type *HTT* and a mutant *HTT* allele. Both normal alleles usually contain no more than 27 CAG repeats while a single mutated allele in HD possesses more than 39 repeats [3,4], and there exists an inverse correlation between CAG repeat length and HD onset age [5–7]. The products of mutant *HTT* including mRNA, protein, and RNA-translation may generate cytotoxicity and eventually lead to HD, in which mutant HTT (mHTT) with a toxic gain-of-function has been considered to be the major cause of the disease although the molecular mechanisms of HD pathogenesis still remain unclear [1,8,9]. By contrast, wild-type HTT (wtHTT) may play a role in normal adult function. It would be very challenging to rely on traditional chemical approaches to rectify most of these abnormalities. However, as a monogenic disease, eliminating mutant *HTT* products serves as the most straightforward therapeutic option. The toxic function of mHTT combined with the performance of dominant and complete penetrance would make the approaches of allele-selective reducing

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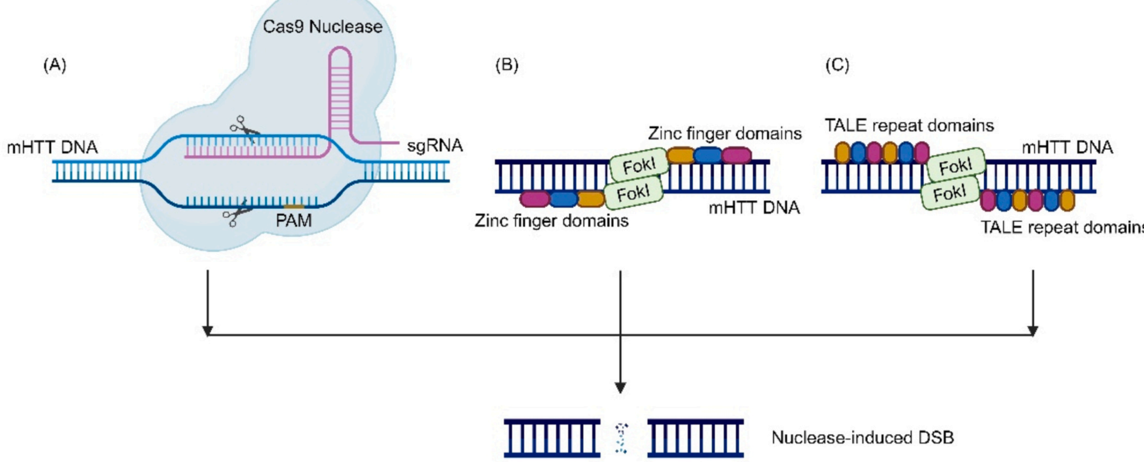

Fig. 1. The mechanism of CRISPR/Cas9, ZFN and TALEN. (A) The CRISPR/Cas9 system consists of a single guide RNA that recognizes target DNA sequence based on a complementary base-pairing rule and the Cas9 nuclease that cleaves mutant DNA containing PAM. The PAM is located in close proximity to the target site in the nontarget strand and plays an essential role in initial recognition and cleavage. (B) Each ZFN is composed of two domains: zinc finger domains for specifically recognizing the mutant gene and the cleavage domain of *Fok*I. (C) Like ZFN, TALE repeat domains are fused to *Fok*I, which constitutes TALEN. Ultimately all of them will produce a nuclease-induced DSB. Abbreviations: CRISPR, clustered regularly interspaced short palindromic repeats; PAM, protospacer adjacent motif; mHTT DNA, mutant huntingtin DNA; sgRNA, single guide RNA; ZFN, zinc-finger nuclease; TALEN, transcription activator-like effector nuclease; TALE, transcription activator-like effector; DSB, double-strand break.

mHTT become a major therapeutic advance in comparison with allele non-selective approaches that indiscriminately inhibit both mHTT and wtHTT for HD treatment [8].

The strategies for lowering mHTT currently under investigation can be classified into two main kinds: allele non-selective and allele-selective approaches using single nucleotide polymorphisms (SNPs) and haplogroup analysis [10]. Given the earlier discovery of allele-non-selective approaches and their rapid development, it is necessary to review the advantages of allele-selective approaches over allele- non-selective approaches. Therefore, this review will begin with the comparison between allele-selective and allele-non-selective approaches followed by an overview of recent progress in allele-selective approaches along with the resulting agents with a focus on pre-clinical and clinical researches conducted on exclusively targeting mutant *HTT* or mHTT in adult HD animals and humans because the therapeutic approaches in clinical trials are designed for HD adult. Furthermore, the existed issues and challenges of allele-selective approaches will also be attempted to discuss including administration, animal models, as well as drug delivery.

2. Allele-selective and allele non-selective approaches

Allele non-selective approaches was initially discovered as straightforward methods to reduce mHTT expression. However, the high similarity of protein sequences between mHTT and wtHTT makes it difficult to effectively differentiate them, and thus results in mHTT lowering concomitant with wtHTT reduction. These approaches has been extensively used to disease-modifying therapeutic research for HD, and the resulting candidate agents are assessed and a few of them have exhibited inspiring outcomes for adult HD in clinical phase [11–15]. Emerging evidences reveal that wtHTT may function as an important scaffold protein in cellular processes such as vesicular trafficking and selective autophagy, and it is essential to maintain neuronal homeostasis and survival in healthy adults except its role in development[16–20]. The failure of tominersen targeting both mHTT and wtHTT in the phase III clinical trial due to unfavorable risk-benefit profile provides support for the essential role of wtHTT in adult brain because it is conceivable that non-selective reduction of HTT might have been a contributing factor [21]. A recent report suggests that a novel role of wtHTT as an essential regulator of short- and long-term plasticity in the adult hippocampus [22]. Moreover, numerous findings indicate that early loss of wild-type *HTT* or reduction of wtHTT levels would contribute to HD pathophysiology in the adults, and even produces more enduring deficits than mHTT [23–28]. Since there are concerns that wtHTT reduction may outweigh the potential benefit of inhibiting the total HTT containing wtHTT and mHTT, further investigations at the preclinical level are required to understand how wtHTT reduction may impact health outcomes following non-selective HTT lowering.

The toxic mHTT is predominantly accountable for the characteristic pathophysiology and observable symptoms of Huntington's disease. Consequently, allele-selective reducing or eliminating mHTT without interfering wtHTT level may offer an ideal strategy for addressing the proximate cause of the disease. Additionaly, genetic studies show a higher percentage of carrier with heterozygous allele on the chromosome pairs compared to the individuals bearing rare homozygotes for HD [2,29,30]. Accompanied by technological advancements in gene therapy containing antisense oligonucleotide (ASO), small interfering RNAs (siRNA), and genome editing, the allele-selective strategy is made technically feasible, and may be a more efficacious and safe strategy than the non-selective strategy [31], and several therapeutic agents selectively aimed at mutant *HTT* mRNA have been brought to clinical trials [32]. In principle, allele-selective strategies could be achieved by blocking mHTT production pathway, or by inactiving and degrading mHTT. The technologies used in the development of agents are mainly grouped into three main classes: (1) targeting mutant *HTT* DAN: mutant DNA sequences can be targeted by clustered regularly interspaced short palindromic repeats (CRISPR)/Cas, or by an inactive zinc-finger nuclease (ZFN) binding to the mutant *HTT* transcription initiation region to prevent its transcription, or by a transcription activator-like effector nuclease (TALEN) recognizing and binding to specific DNA sequences to shorten the expanded CAG repeat tract; (2) targeting mutant *HTT* mRNA: its level could be individually reduced or its translation be specifically inhibited by chemically modified oligonucleotide technologies including shRNA, siRNA, and ASOs; (3) targeting mHTT: post-translational modification of mHTT, including reducing its aggregation or enhancing its clearance, as a pharmacological approach by small molecule agents acted via either ubiquitn proteasome system (UPS) pathways or autophagy pathways. With the exception of genome editing approaches, these techniques will reduce but not completely block mHTT production. Therefore, the term "mHTT-lowering" is the preferred term for this class of potential therapies.

3. Mutant *HTT* DNA-based approaches

3.1. CRISPR/Cas9 system

The allele-selective approaches to targeting only m*HTT* DNA can be implemented by gene editing techniques that have demonstrated successful reductions in mHTT protein levels in animal models and clinical trials [32,33]. These techniques focus on the targeted manipulation of specific DNA sequences to induce double-strand breaks in the desired genomic regions followed by a modification of the target gene or replacement of part of the DNA sequence with a donor sequence for precise repair of the break. Among the widely utilized gene editing technologies are ZFNs, TALENs, and CRISPR/Cas9 system.

The CRISPR/Cas9 system is composed of Cas9 nuclease and a synthetic single guide RNA (sgRNA) [29] (Fig. 1A). Case 9 is guided by specific sgRNA to target specific region of DNA. Importantly, the target DNA sequence must contain a protospacer adjacent motif (PAM) sequence required by Cas9 to recognize and bind its target DNA properly [34], and thus SNP variants-derived PAM recognition might be exploited to differentiate alleles for allele-specific CRISPR-Cas9 editing [35]. Shin and Montey proposed the concept of an allele-specific CRISPR-Cas9 strategy to inactivate the mutant *HTT* by excising a region that is important for gene expression [36,37], and the allele-selective silencing mutant *HTT* has been achieved by designing a sgRNA flanked by the specific PAM that is present only within the mutant allele in cell and animal models [33,38,39]. Based on SNP variations that are different between mutant and wild-type *HTT*, Shin et al. used two CRISPR/Cas9 sgRNAs to simultaneously target two mutant-specific PAM sites generated by PAM-altering SNPs on the mutant chromosome to selectively inactivate mutant *HTT* via eliminating the promoter region, transcription start site and the expanded CAG repeats of mutant *HTT* allele in HD patient-derived fibroblasts [36]. Subsequently, a similar approach was tested in HD patient-derived fibroblasts and BACHD mouse models [37,40]. To decrease the risk of off-targeting that emerged in the usage of dual target sites in the CRISPR/Cas9 approach based on the PAM-altering SNPs, Shin et al. used a single gRNA to selectively inactive mutant *HTT* through nonsense-mediated decay (NMD) [41]. The high applicability of the mutant allele-specific editing using PAM-altering SNPs in the HD patient population was identified through the analysis of phased genotypes of 8543 HD subjects with European ancestry, and its efficacy was validated experimentally using two independent patient-derived induced pluripotent stem cells (iPSCs) lines [42]. Efforts continue to be concentrated on enhancing PAM specificities, the Aligning Patients to Cas (AlPaCas) platform that is used freely has been developed and provides screening tools for sequence-based identification and structural analysis of PAMs derived from SNP variants [39]. The efficacy of the aforementioned CRISPR/Cas9 approaches based on PAM-altering SNPs haplotype-targeting relies on the specific PAM recognition by Cas9 constraints, and thus significantly limits the targetable SNPs. This limitation has been overcome by a paired Cas9 nickase approach that can target variants of naturally recognized PAMs, which effectively removed enlarged CAG repeats in HD patient-derived cell lines including fibroblasts, neurons and astrocytes, and showed the specificity of mutant *HTT* and nearly undetected off-target effects [43,44].

Recently, the feasibility of gene correction and replacement in CRISPR/Cas9 has been also explored for HD treatment. Xu et al. seamlessly corrected the mutant *HTT* with the combination of CRISPR/Cas9 and piggyBac transposon, which ameliorated HD-mediated phenotypic defects in iPSCs derived from HD patients [45]. Yan et al. well demonstrated that the replacement of 150 CAG expansion with a normal 20×CAG can result in the obvious mHTT reduction and movement ability improvement in HD knock-in pig model by single durable intracranial or intravenous injection of CRISPR/Cas9 and corresponding adeno-associated virus (AAV) delivery system [46].

Although the recent discoveries exhibit encouraging, several relevant drawbacks should be addressed before CRISPR can be implemented in clinics. Viral vectors, the commonly used delivery system in CRISPR/Cas9, have disadvantages including small packaging capacity, restricted distribution within the brain, off-target effects induced by indefinite production, immunotoxicity, and risks of insertional mutagenesis [47,48]. Therefore, efforts have been undertaken to address these constraints, such as the total removal of AAV-encoded sequences contributing to expansion in packaging size, low immunogenicity, and cytotoxicity [49]. KamiCas9, a self-destructing CRISPR/Cas9 system designed by Déglon's team, inhibited sustained expression of the Cas9 protein and reduced collateral damage [50,51]. The off-target effects/-mutations could be mitigated by modified gRNA or Cas9 [52] and identified by computational tools and experiment validation.

3.2. ZFN

ZFNs, distinguished from CRISPR/Cas9, offer distinct advantages such as reduced packaging requirements and expanded selection of DNA cleavage sites due to the absence of a PAM motif [53]. However, they are also linked to potential off-target effects and cytotoxicity concerns [54]. Considerable time and high costs would be involved in developing and optimizing ZFNs. In ZFN, the specificity of DNA-binding domains of zinc finger proteins (ZFP) guides the cleavage domain of the *Fok*I endonuclease to generate a double-strand break (DSB), while ZFP without nuclease activity can lower the gene expression levels by simply selective binding to DNA without modifying it [8] (Fig. 1B). In summary, HD researches adopted ZFN and ZFP strategies respectively causing mHTT cleavage and repression.

Mittelman et al. first discovered that ZFN could effectively eliminate extended CAG repeats in human and rodent cell lines associated with HD. Further studies concentrate on designing ZFP structures since current ZFNs cannot satisfy varying DNA, particularly those involving large quantities of DNA [55]. Garriga-Canut et al. reported 95 % reduction of mHTT levels and alleviation of rotarod deficits and clasping following striatal injection of CAG-targeting AAV-ZFP in R6/2 mice [56]. To minimize reliance on the poly-CAG tract and reduce allele skewing, Zeitler designed a new class of zinc finger-based transcriptional factors (ZF-TFs), which repressed HD-causing alleles by >99 % while preserving wild-type by more than 86 % in vitro. More encouragingly, these factors were well tolerated for at least 9 months in the mouse brain, suggesting that the development of zinc finger technology may provide a significant roadmap for future HD treatments. [57].

3.3. TALEN

TALEN is a structure of fusion protein that possesses transcription activator-like effector (TALE) domain for specific recognition, which fused to a *Fok*I nuclease for generating a DSB [58] (Fig. 1C). Compared with ZFN, TALEN is more affordable and has relatively well-defined target specificity along with cleavage efficiency. However, the difficulty in complicated protein engineering [59] and packaging in certain viral vectors (e.g., lentiviral vectors) restrict the application of TALENs for HD genome editing [60].

Similar to ZFNs and ZFPs, TALENs and TALEs are also employed in HD modification for cleavage and repression purposes. Richard et al. successfully induced the TALEN in yeast, achieving the contraction of expanded CAG/CTG repeats with nearly 100 % efficacy, while observing no increase in mutation rate or genomic rearrangement [61]. Fink et al. published the only report about TALEN associated with HD. This study utilized two distinct methods: a SNP-based TALE paired with a transcriptional repressor to suppress the expression of mutant *HTT* without nuclease activity, and expended CAG repeats binding TALENs with

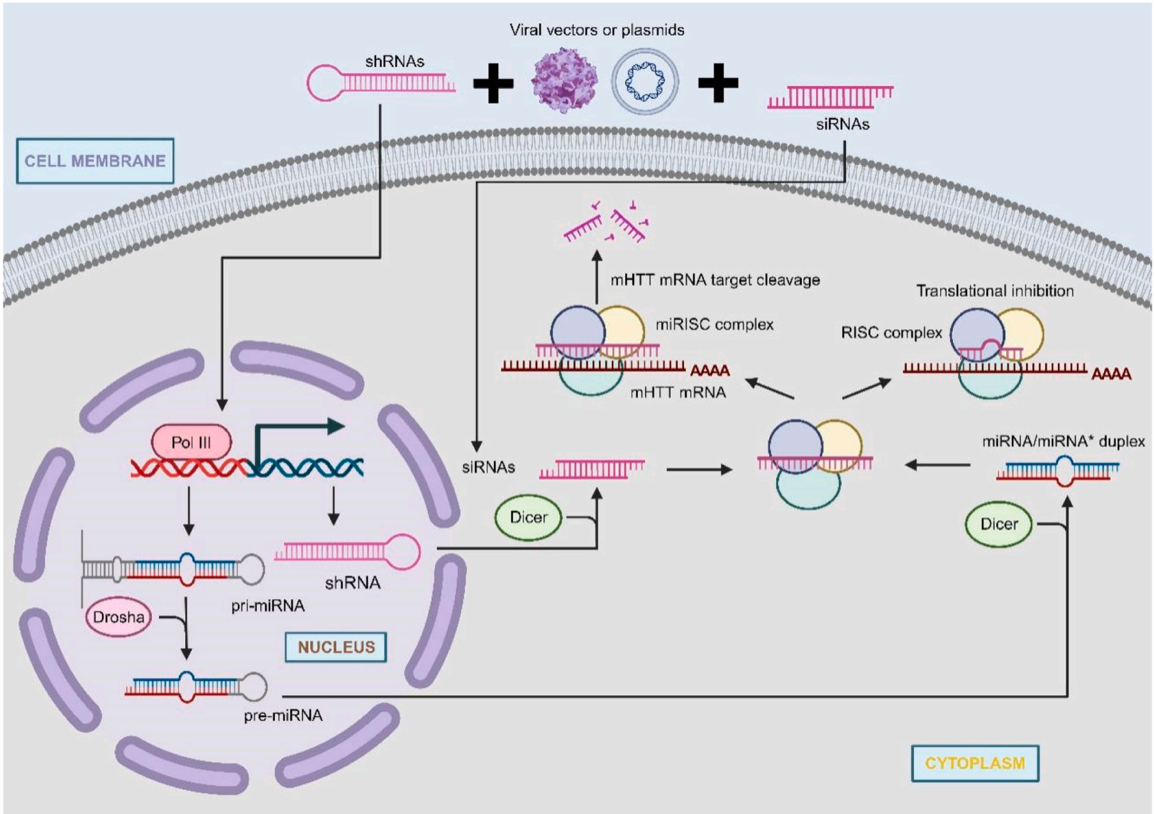

Fig. 2. The mechanisms of RNAi. In the nuclear, pre-miRNA is formed after the activity of Drosha enzymes on pri-miRNA that is transcribed by poly III. After nucleus transportation and the processing of Dicer, it is acted upon by Dicer to generate a mature miRNA/miRNA* duplex structure. The role of miRISC complex degrades the unstable miRNA* strand and the mature strand is utilized for subsequent translation inhibition. With directly introduction to the cytoplasm or plasmids and viral vectors assistance, synthetic siRNAs entered the cytoplasm. shRNAs can be also delivered by vectors and transit to the siRNA pathway by Dicer. Both of them activate the RISC complex for selecting the guide strand for mHTT mRNA target cleavage. Abbreviations: RNAi, RNA interference; miRNA, microRNA; pri-miRNA, primary microRNA; pre-miRNA, precursor microRNA; miRISC, microRNA-induced silencing complex; siRNA; short interfering RNA; shRNA, short hairpin RNA; RISC, RNA-induced silencing complex.

cleavage activity. Although a reduction of aggregation was observed in HD fibroblasts, the efficiency remained suboptimal [62]. To data, research on the application of TALENs in HD animal models is still at an exploratory stage, no result has been demonstrated on the use of TALEN in HD animal models, which may be attributed to the absence of established delivery systems, reliable data on toxicity, or information about the impact on phenotype available for further discourse [63].

3.4. Compounds targeting DNA

A chemical perspective encourages researchers to develop synthesized molecules that target mutant *HTT* DNA. Nakamori et al. synthetized naphthyridine-azaquinolon (NA), which induces guanine-adenine (G-A) mismatch and extrudes two cytosines from the CAG harpin. This compound resulted in the contraction of the striatum in 6-week-old R6/2 HD mice with 150 CAG repeats [64]. Its derivative NBzA also displayed a comparable binding affinity for $(CAG)_n$ repeat DNA [65]. These mutant *HTT* DNA-targeting small molecules still develop slowly though they show the promising future of the HD treatment. Future efforts should thoroughly consider the feasibility of introducing nucleases or transcriptional modulators and integrating them into endogenous systems. [66].

4. RNA-based approaches

4.1. RNAi

RNA interference (RNAi) technology provides a versatile means to modulate target gene expression, which has become an adaptable and powerful therapeutic strategy, and the resulting RNAi-based drugs

have been approved for clinical use [67]. For allele-selective HD treatments, RNAi approaches are utilized to inhibit mutant *HTT* mRNA translation or degradate mutant *HTT* mRNA. According to their origins, structures, and associated effector proteins, RNAi molecules are mainly divided into three classes: microRNA (miRNA), short interfering RNA (siRNA), and short hairpin RNA (shRNA), converging into RNA-induced silencing complexes to achieve post-transcriptional gene regulation (Fig. 2).

4.1.1. miRNAs

miRNAs are short, non-coding RNAs derived from long stem-loop structures, which results in imperfect complementary and translation silencing rather than degradation [68]. The biogenesis of miRNA is initialized by the transcription of microRNA (miRNA) genes. Following endonucleolytic cleavage, nucleus transportation, and the processing of Dicer, the transcript is converted into mature miRNA/miRNA* duplex. When loaded on microRNA-induced silencing complex (miRISC), one miRNA strand, the passenger strand is discarded while the other strand, the guide strand is selected and directs miRISC to bind with complementary mRNA and suppress the translation [69,70]. In HD animals,

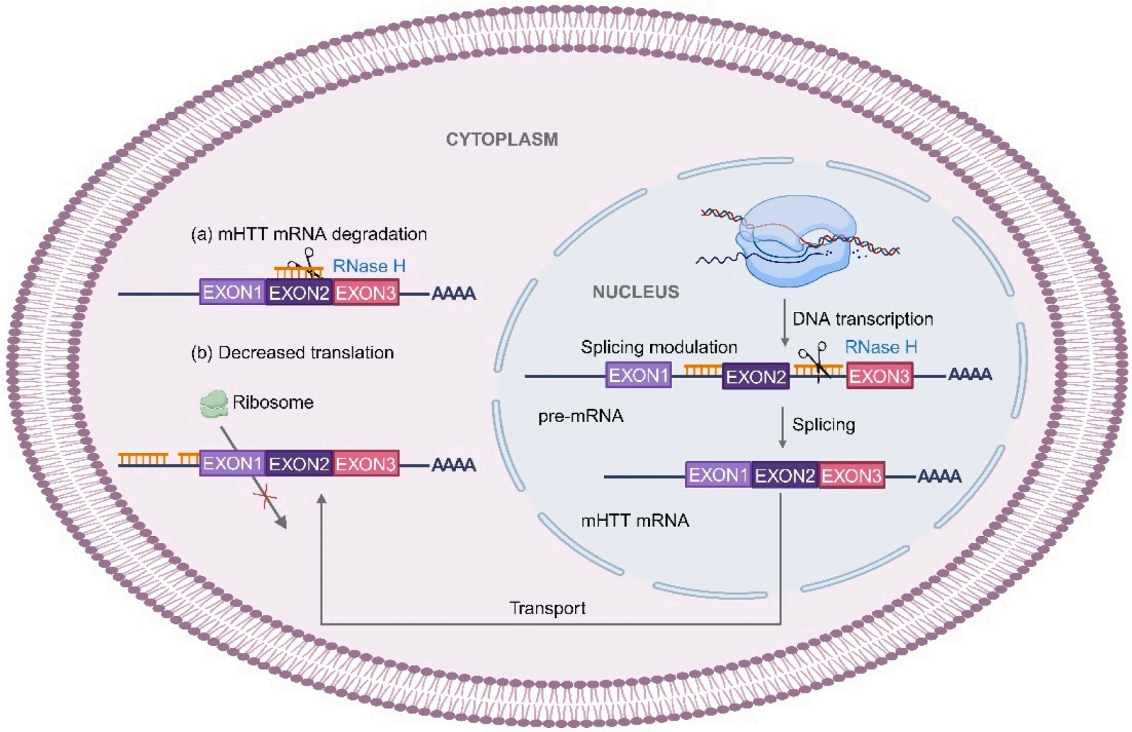

Fig. 3. The mechanisms of ASO. Given therapeutic ASO chemical design and target, they exhibit their effects by several different mechanisms of action. In the nucleus, ASOs can bind on pre-mRNA intron/exon junctions and recruit RNase H nuclease for degradation. In the cytoplasm, they can also (a)activate RNase H and induce the cleavage of the mHTT mRNA or (b)prevent the physical assembly of the 40S and 60S ribosomal subunits onto the mRNA sequence or interaction with crucial sequence on mRNA. Abbreviations: ASO, antisense oligonucleotide; pre-mRNA, precursor messenger RNA.

artificial miRNAs are usually delivered into the nucleus via AAV vectors and, effectively mimicking the function of endogenous miRNAs over an extended period [71].

Monteys et al. used SNP-targeting artificial miRNAs and found allele preference in HEK293 cells as well as in HD mouse model. Notably, it was also found that when AAV-miRNAs increased, their allele specificity got reduced [72]. This may suggest that the endogenous RNAi pathway becomes oversaturated due to competition between the production of permanent artificial miRNAs and endogenous systems.[73]. To optimize the expression of a microRNA and enhance the allele selectivity, pri-miRNA scaffolds present an alternative option. Miniarikova et al. selected the most effective pri-miRNA scaffold towards SNPs in vitro and evaluated the neuronal dysfunction and immune response in Hu128/21. The promising result was that the level of DARPP-32 was lowered and no signs of astrocytes or microglia were detected. The passenger strand was also removed considering that the excessive production is associated with off-target effects and toxicity [74,75]. Additionally, pri-miRNA scaffolds could also target the CAG repeat and reduce the level of mHTT by 50 %. Furthermore, these scaffolds exhibited good tolerability over a 20 weeks in YAC128 mice. [76].

miRNA is an attractive approach for the reduction of mHTT levels and the exploration of the mechanism action of artificial miRNAs will be beneficial for improving selective silencing ability [77]. However, accumulated evidence revealed that miRNA has off-target silencing effects and is strongly correlated to the partial complementarity in 3′UTR of unintended genes with the seed region of miRNA [78]. It can be minimized by rigorous filtration for both guide and passenger strands of the RNAi effector molecules and this approach has been validated.

4.1.2. *siRNAs*

Different from miRNAs, siRNAs are mainly regarded to be exogenous in origin and derived from fully complementary double-stranded RNAs (dsRNAs). With direct introduction to the cytoplasm or plasmids and viral vectors assistance, synthetic siRNAs bypass this nuclear step and activate the RNA-induced silencing complex (RISC) for excising the passenger strand and leaving the guide strand that incorporates into the RISC [79,80]. The perfect complementarity induces mHTT mRNA degradation. The main difficulties of siRNAs approaches are durability and limited diffusion. When directly administrated into the brain, siRNAs show effective silencing ability for only a short period and remain spatially confined to the area surrounding the injection site [81].

Both CAG-targeting and polymorphism-based methods such as SNPs or a deletion of the *HTT* gene have shown a preference in reducing mHTT levels [82–85]. Cooperated Efforts have been devoted and cooperated[89] to better discrimination between mutant *HTT* and wild-type *HTT* mRNA: (a) Chemical modifications developed in ASOs or new modifications that are never seen in other RNA therapies [86,87]. (b) Abasic substitutions and mismatches [88–92]. (c) varying siRNAs length and the location of modification.[87] (d)single-strand siRNAs (ss-siRNAs) [93]. Meanwhile, the probability of applying allele-selective siRNAs for the majority of the HD patients has been evaluated and increased [94,95], which is beneficial for the HD treatment in the future.

4.1.3. *shRNAs*

shRNAs are artificial RNA molecules with hairpin or loop-like structures. Intriguingly, many studies have compared shRNAs with siRNAs or miRNAs [96,97], revealing that shRNAs are more efficient in gene silencing than miRNAs and have lower off-target than siRNAs. Transcribed on plasmids or viral vectors, shRNAs can induce long-term expression while preserving the potency and allele-specificity of the original siRNA. They are transported from the nucleus like miRNAs and transit to the siRNA pathway by Dicer [98]. The expression of shRNA is difficult to predict and control, as the Dicer cleavage site may make one or two-nucleotide discrepancies between the expected siRNA and the produced siRNA, although the process of selection and mutant *HTT* mRNA cleavage is the same as siRNAs [99].

Drouet et al. addressed that SNP-shRNAs not only reduced mutant *HTT* RNA but also reestablished axonal transport as well as restored vesicular function in human embryonic neural stem cells [100]. A recent study also found the result of silencing mHTT levels with CAG-targeting shRNAs [101]. The major concern is toxicity associated with high-dose shRNAs after the delivery of shRNA into the cell. Data indicate that 4 months after instriatal injection of both control and HTT-targeting AAV-delivered shRNAs, neuronal cells died with increased inflammation evidenced by microglia in the striatum. These toxicities are thought to be secondary to the specific shRNA introduced and may be associated with the absolute amount of siRNA generated. Notably, some AAV-delivered shRNAs did not exhibit toxicity, and the production levels of nontoxic shRNAs were found to be lower in comparison to those of toxic shRNAs. [102].

4.2. ASOs

ASOs are short oligonucleotides that specifically target cellular mRNA through complementary base pairs, exhibiting minimal side effects. In the past few years, significant efforts have been made to develop ASOs and two therapeutic agents were approved by the FDA for the treatment of Duchenne muscular dystrophy (DMD), and spinal muscular atrophy (SMA). This progress enhances the prospects for similar therapeutic modalities in various neurological disorders, like HD [103]. Unlike RNAi, ASOs are considered to recognize their targets independently, without the assistance of auxiliary proteins and depend on their ability to recognize, and hybridize with complementary RNA sequences. Upon binding to the mutant *HTT* mRNA, ASOs can induce degradation of the target mRNA in either the nucleus or the cytoplasm via the recruitment of RNase H nuclease. A second common antisense mechanism, mHTT translation inhibition, occurs through steric blocking translational machinery from assembly or interacting with important RNA sequences such as initiation codons [104,105](Fig. 3).

Multiple ASOs, targeting either SNPs or expanded CAG tract, have demonstrated their allele-selectivity in vitro and in vivo [106,107]. Nevertheless, during the early development of ASOs, original ASOs were unstable and susceptible to fast degradation by nucleases. To enhance their stability, affinity, selectivity, potency and decrease unwanted toxicities, several modifications to ASOs have been explored and implemented to ASOs, increasing their potential as a HD therapeutic agent [108,109]. Here, we will briefly describe tremendous progress in allele-selective chemically modified ASOs in the HD therapeutics field.

Modifications to the phosphate backbone contain phosphorothioate (PS), phosphorodiamidate morpholino oligomer (PMO), and peptide nucleic acid (PNA). The use of the PS backbone, in which a sulfur atom replaces a non-bridging oxygen atom of the phosphate group, substantially enhanced resistance against RNase H degradation, and prolongs the half-lives of ASOs in serum and cerebrospinal fluid (CSF) without obvious non-specific toxic side effects [110]. The issues about the structure of PS ASOs such as length, number and position of chemical modifications and the combination of other modifications have been thoroughly discussed in vitro and in vivo studies [111–113]. The tolerability and drug-like properties of ASO candidates were also evaluated in Hu97/18 mice [114]. A disadvantage of the PS backbone is that it may disturb normal protein functions causing toxicity and activating the complement system contributing to immunogenicity [115]. Other backbone modifications such as PMOs, involve a sugar-phosphate substitution where the sugar moiety is replaced by a morpholine ring and the charged phosphodiester linkage with uncharged phosphorodiamidate linkage. Like the PS oligonucleotides, PMOs are highly resistant to nuclease and protease degradation. They do not support RNase H nuclease activity and are primarily used for translation inhibition [116]. Sun and colleagues designed a series of PMOs that were capable of selectively targeting mHTT mRNA. Their findings demonstrated effectiveness in significantly reducing the mHTT levels of HD patient-derived fibroblasts as well as the N174–82Q transgenic mouse models and improving mHTT-induced toxicity in neuronal cultures. Moreover, intracerebroventricular (ICV) injection of PMOs resulted in an improvement of depressive-like phenotype in the HdhQ7/Q150 mouse model [117]. However, these authors chose HdhQ7/Q150 mice at 6,8,10 months of age and only a limited behavior analysis could be performed since they exhibited very little phenotype at 12 months of age. Given that the models are significant in the research of HD, the choice of models should consider their disadvantages and reflect the question of interest in refining existing PMOs and developing novel therapies for treating HD [118,119]. The PNAs are generated by the replacement of the sugar- phosphate backbone with a peptide backbone composed of *N*-(2-aminoethyl) glycine units. Similar to PMOs, PNAs are neutrally charged, cannot support RNase H activity, and have a high binding affinity for RNA. By varying PNA length and peptide conjugation or introducing chemical modification, Hu et al. increased selectivity in the HD patient-derived cell lines [84,120]. Friedlander et al. demonstrated that PATrOL™-enabled DNA-directed PNA NT-100D was able to cross the BBB and showed good tolerance at pharmacologically active doses in the R6/2 transgenic mice. The observed improvement of functional rescue in a dose-dependent manner and the absence of changes in wtHTT and NfL levels suggested the selectivity in reducing mHTT significantly throughout the brain without inducing neurotoxicity [121]. One of the shortcomings of PMOs and PNAs is that they are rapidly eliminated via urine excretion due to poor cell permeability and water insolubility, which needs repeat administration and thus increases side effects [122].

In addition to the phosphate backbone, ASOs can be modified at the 2′ position of the ribose sugar, thereby enhancing their pharmacological effects. Examples such as 2′-O-methyl (2′-OMe), 2′-O-methoxyethyl (2′-MOE), 2′-fluoro, S-constrained-ethyl (cEt), and locked nucleic acid (LNA) have shown enhanced nuclease stability (with 2′-fluoro being the exception), decreased toxicity potential, and increased hybridization affinity to mutant *HTT* mRNA. 2′-OMe ASOs have a methyl group at its 2′ position added to the ribose. For 2′-MOE, the hydrogen atom at the 2′-position of the sugar is replaced by an O-alkyl group. The 2′-fluoro, similar to the 2′-MOE modification, substitutes the 2′-hydroxyl group with a fluoro group and exerts superior binding affinity compared to most other 2'-modifications [123]. cEt and LNA have 2′-4′ ethyl chain and a methylene linkage respectively. They have similar binding affinity and provide increased potency compared with 2′-MOE ASOs. But cEt appears to have a more favorable toxicity profile than LNAs. A few comparisons suggest that LNA is more soluble and often stronger to hybridize RNA than PNA and 2′-MOEs seem to be safer than LNAs in experiments [123–125]. These chemistries rely on translation inhibition of the target mRNA rather than RNase H-mediated degradation and individually or cooperatively preferentially targeted the mutant *HTT* transcript in vitro studies [84,106,108,109,120,126,127]. Additional modifications such as zip nucleic acid (ZNA) and [bis-o(aminoethoxyl) phenyl] pyrrolocytosine (PhpC) are respectively conjugated into ASOs and have recently shown good promise in HD preclinical trials [128, 129]. Further animal experiments will be required to explore pharmacodynamics and determine the effectiveness of translation to human therapy.

In terms of stereochemistry, conventional PS ASOs consist of a mixture of multiple stereoisomers, some of which have therapeutic effects, while others are less beneficial or could even contribute to toxic outcomes [130]. Compared with these stereorandom ASOs, stereopure ASOs exhibit enhanced stability PS-linkages, improved specificity, and reduced immunogenicity although it is still debated whether stereopure ASOs can be more potent than stereorandom ASOs [103,131]. Based on this principle, the pharmaceutical company Wave Life Sciences developed two stereopure ASOs, WVE-120101 (NCT03225833) and WVE-120102 (NCT03225846), which are designed to modulate activity and selectively target mutant *HTT* mRNA. Each of them directs against SNPs generally appearing on the mutant allele and has entered the clinical trial stage [132,133]. Recently, both drugs respectively have

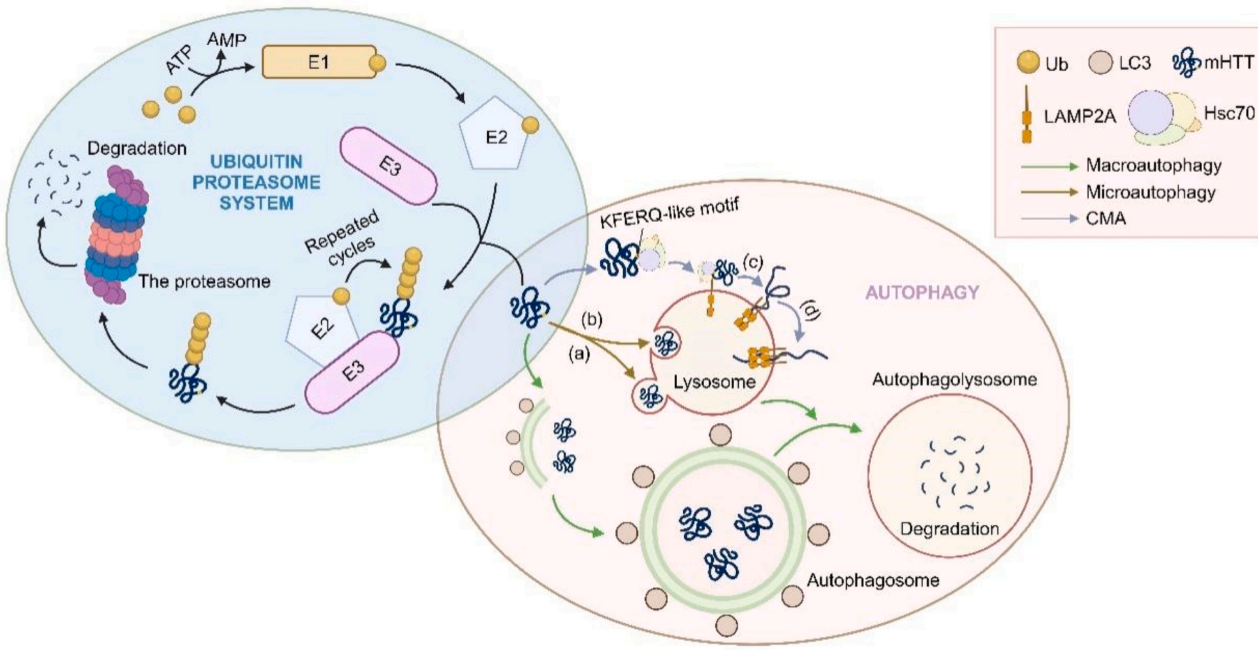

Fig. 4. The mechanisms of autophagy and UPS. Autophagy contains macroautophagy, microautophagy and CMA. In macroautophagy, mHTT are sequestered by the expanding phagophore, leading to the formation of the autophagosome. LC3 is involved in the assembly of the isolation membrane and assists to form the autophagosome. The autophagosome subsequently fuses with lysosome and eventually mHTT are degraded by hydrolases in the autophagolysosome. In microautophagy, mHTT are directly taken up by the (a)protrusion or (b)invagination of the lysosome, followed by exposing mHTT to hydrolases for degradation. In CMA, the CMA-targeting motif (KFERQ-like) in the mHTT is recognized by cytosolic chaperone Hsc70. This delivers it to the surface of the lysosomes where it binds to LAMP-2A. Then LAMP-2A (c)turn into high order multimeric complexes, which is required for subsequent (d)mHTT translocation and degradation. In UPS, the Ub-activating enzyme E1 forms a covalent bond with Ub in an ATP-dependent process. The Ub-conjugating enzyme E2 subsequently transfers Ub from the E1 to itself. The Ub ligase E3 recognizes mHTT and transfers Ub from the E2 to mHTT. After repeated cycles, the conjugation of mHTT and Ubs is recognized by the proteasome which subsequently degrades mHTT to peptides and recycles the ubiquitin. Abbreviations: mHTT, mutant huntingtin; CMA, chaperone-mediated autophagy; LC3, light chain 3; Hsc70, heat shock cognate 70 kDa protein; LAMP-2A, lysosome-associated membrane protein type 2 A; UPS, ubiquitin proteasome system; Ub, ubiquitin.

gone through phase I/Ia clinical trials, PRECISION-HD1 and PRECISION-HD2 in people who manifest early phenotypes of HD to evaluate the safety and tolerability of the drugs. Early data for these trials including dosages of 2, 4, 8 and16 mg has been made public through a press release. At these dosages, most of the patients in the experimental group and the control group experienced adverse events that not thought to be serious [134]. Moreover, a dose-dependent mHTT in CSF decreased by 12.4 % and no change of CSF NfL (neurofilament light chain) was observed, demonstrating that the axons were not damaged and preliminary effects were passable after evaluation. The comparatively slight decrease in CSF mHTT has revealed that the company should test higher doses of both agents. Therefore, a 32 mg trial was started to assess whether dose increment could safely reach the mutant *HTT* knockdown level considered effective. However, both studies were announced to be discontinued due to failure to constantly and significantly reduce mHTT in CSF and transient serious adverse events of the HD patients. The main problem in these ASO clinical trials is the lack of appropriate animal models with the related SNPs when allele-selective stereorandom ASOs were developed [135], and thus it was unable of convincingly validate the pharmacokinetic and pharmacodynamic effects of these molecules. Nonetheless, it is still reasonable to be optimistic about what this strategy may achieve because this is a different problem from that seen with tominersen, which did lower levels of mHTT but did not seem to slow the progression of the disease [136]. Now the company plans to launch a new ASO, WVE-003 (NCT05032196), which targets a different SNP from the previous trials. In light of the promising preclinical data demonstrating mHTT-lowering effects in motor neurons and BACHD mice, Wave is working hard in the direction of WVE-003 and has initiated a phase I trial sometime in 2021. The safety and tolerability, pharmacokinetic and pharmacodynamic assessments in plasma and CSF, together with fluid biomarkers and the composite Unified Huntington's Disease Rating Scale (cUHDRS) will be assessed and receive attention [136].

4.3. *Compounds targeting RNA*

The researches in RNA biology underscore that RNA plays a central role in all biological systems. Numerous small molecules and CRISPR-based systems have been discovered and selectively bound to repeat-containing RNAs [137]. Previous studies have shown chromenone-based molecules pyridocoumarins are specific for r(CAG) exp RNA and decreased mHTT production in a dose-dependent manner in cellular models of HD [138]. Myricetin was also found to selectively interact with 5'CAG/3'GAC mHTT mRNA sequence and therefore stop both the translation of mutant HTT protein and the sequestration of the HD-related muscleblind-like 1 (MBNL1), resulting in the mitigation of mitochondrial dysfunction and oxidative stress, along with motor deficits in 3-nitropropionic acid (3-NP)-induced HD rats [139]. But there are inherent risks in showing a pronounced discrimination between target and off-target sites owing to the intercalation and stacking [137]. Furthermore, several compounds based on the structure of myricetin were screened by the National Cancer Institute (NCI) [140]. Among them, CP13 exhibited higher affinity and less toxicity compared with myricetin. Recently, mRNA splicing modulators like CRISPR-Cas13d have been reported to decrease the accumulation of mHTT with limited off-target effects on both HD mouse and human striatal neuron transcriptome. When packaged in a constitutive AAV vector, CRISPR-Cas13d attenuated striatal atrophy and improved motor coordination lasted for at least 8 months in HD mice [141,142].

These molecules hold promise, providing new avenues for HD therapeutic intervention. However, low information content and low pocket quality lead to trade-offs in drug-likeness and potency between mHTT mRNA-targeting molecules [137]. In addition, the consideration of RNA motifs with structural sophistication and detailed characterization of small molecule-RNA interactions have been largely neglected [137, 143]. Further studies are hoped for full and deep understanding of the biological effects and side effects before human trials with the support of a wide variety of screening strategies.

5. Protein-based approaches

Although gene therapies such as CRISPR/Cas9, RNAi, and ASOs effectively target DNA or RNA levels and can suppress mHTT selectively in vitro and animal models, these techniques are devoid of the target's druggability and are often challenged by nucleic acid delivery issues, irreversible genome changes and unwanted off-target effects [144]. Therefore, an avenue exploited for HD is the induction of new protein silencing tools for reducing and degrading targeted protein aggregations. The intracellular proteolytic machinery and antibodies that selectively recognize domains of mHTT have been a recent focus and shown to ameliorate toxicity in various HD models.

Two main pathways for protein breakdown are the autophagy-lysosomal system and ubiquitin proteasome system (UPS). In mammals, three main forms of autophagy have been identified, macroautophagy, microautophagy, and chaperone-mediated autophagy (CMA). Macroautophagy is the most comprehensively studied type of autophagy. The initial step of macroautophagy includes the formation of a phagophore, a lipid bilayer membrane structure for isolation. The lunate-shaped phagophore enlarges to swallow intracellular protein aggregates, thereby closing off to form the double-membraned an autophagosome. The lysosome comes and fuses with the autophagosome to turn into autophagolysosome and degrade the inner contents by lysosomal acid proteases (Fig. 4) [145]. In microautophagy, cytosolic contents are directly internalized by the lysosomes via membrane protrusion or invagination, ultimately leading to their degradation within the lysosomal lumen [146]. For a long time, it was generally believed that the autophagy pathway was a non-selective process in which each autophagosome traps and digests bulk cytosol. However, emerging evidence has highlighted selective autophagy machinery could be a possibility [147]. Thus, the discovery of CMA occurred and is motivated by the observation of unique lysosomal internalization different from microautophagy and the identification of proteins responsible for substrate binding, translocation, and uptake [148]. The special requirement for chaperone in the process renamed this type of autophagy as CMA in 2000. During CMA, substrate proteins contain KFERQ-like motifs, which facilitate selective degradation within lysosomes and are recognized by cytosolic chaperone heat shock cognate 70 kDa protein (Hsc70). Once bound to the chaperone, the complex interacts with lysosome-associated membrane protein type 2 A (LAMP2A) on the surface of the lysosomes, promoting LAMP2A assembly into a multimeric complex that facilitates substrate translocation. After unfolding and internalization, the target protein is permitted to reach the lysosomal matrix for degradation.

The UPS is the second main degradation in mammalian cells, mainly functions in the degradation of short-lived, soluble proteins while maintaining cellular homeostasis, and its targets are marked with ubiquitin (Ub) for degradation. First, the E1 activating enzyme uses energy from ATP hydrolysis to form a thioester bond with Ub for activation, the activated Ub is subsequently connected to the E2 conjugating enzyme and ultimately transferred to the substrate mediated by E3-ligase. Through repeated cycles of ubiquitination, a substrate is covalently modified by poly-Ub chains, which are recognized by the proteasome for degradation.

Many small molecules have been identified as stimulators of autophagy by targeting autophagy-related pathways or protein factors participating in the macroautophagy (hereafter referred to as autophagy) machinery. Several compounds including clonidine, minoxidil, and L-type Ca^{2+} channel antagonists have been identified that modulate autophagy through the regulation of cyclic AMP (cAMP)/inositol triphosphate (IP3) and Ca^{2+}–calpain–Gsα pathway, which have been shown to ameliorate phenotypes in HD animal models without obvious effects on wtHTT [149]. HTT can also be subjected to multiple post-translational modifications. The acetylation at lysine 444 enhances the clearance of mHTT via the autophagic-lysosomal pathway, and wtHTT was comparatively acetylated at lower levels. However, it is not clear if this process can be enhanced selectively and requires further validation [150]. Another study indicated that MAP1S, which regulated the autophagy flux through autophagy-related protein light chain 3 (LC3). When HDAC4 that colocalized with mHTT but not wtHTT and MAP1S bind to each other, the stability of MAP1S is compromised, leading to an increased accumulation of mHTT aggregates. Therefore, short peptides or others that inhibit HDAC4 or overexpress the HDAC4-MAP1S interaction domain HBD stabilized MAP1S and restored degradation of mHTT [151]. Pierzynowska et al. reported that genistein, a natural isoflavone, was regarded as an autophagy stimulator in experimental studies and decreased both aggregates and the soluble form of mHTT on HD cellular models. The treatment of genistein can cross BBB and was proven to be safe for long-term use even at high (150 mg/kg/day) doses without affecting wtHTT levels dramatically [152]. Li et al. have demonstrated certain small molecule glues, referred to as ATTECs (autophagosome-tethering compounds) can effectively bind to both LC3 and mHTT, leading to a significant reduction in HD pathologies in vitro and in vivo in fly and mouse models [153]. However, these compounds also contained pan-assay interference compounds ('PAINS') substructures that need to be removed in future optimization [154]. More recently, Long and colleagues also identified a small molecule with good BBB permeability, named J3, that selectively cleared soluble and insoluble mHTT without affecting wtHTT levels in vitro via both UPS and autophagy, which was induced by mTOR inhibition. Nonetheless, when tested in HdhQ140, J3 slightly improved motor functions and lowered total HTT levels in the striatum [155]. Perhaps J3 does not recognize the target in vivo or fails to reach the site of action [156]. The limitation of this experiment is that the tested HdhQ140 mice and wild-type mice were not littermates. Since environmental factors contribute significantly to behavioral responses, caution should be exercised when interpreting the results of the behavior tests [157]. Previous studies have shown that Valosin-containing protein (VCP/p97) colocalized with mHTT and can interact with LC3. This suggests its involvement in neurodegenerative disorders and highlights its significant role in autophagy. Li et al. discovered that gossypol acetate (GA), a traditional Chinese medicine, modulates VCP−LC3−mHTT ternary complex interaction and improves associated motor function deficits in transgenic mouse models of HD [158]. Moreover, Wrobel et al. discovered a small molecule VCP activator, SMER28 could boost autophagosome biogenesis by increasing PI(3)P production and induce autophagy and UPS-mediated degradation of mHTT and its aggregates while preserving the wild-type counterparts [159]. Given that intrabodies could selectively target the pathological protein and have the potential role in neurological diseases [160], a small intrabody peptide SM3 was developed to facilitate the transport of mHTT to lysosomes for degradation. The brain injection or intravenous administration effectively reduced soluble and insoluble mHTT in the brains of HD KI-140Q mice leading to improvements in HD-related neuropathology and motor function deficits [161].

Besides autophagy, the CMA mechanisms also facilitate the design of an engineered fusion peptide comprising two different KFERQ-like motifs and two copies of polyglutamine-binding peptide 1 (QBP1), which interacted with expanded polyQ tracts and has succeeded in reducing neurotoxicity and extending lifespan in R6/2 mouse models via the intrastriatal delivery [162]. Multiple intervention nodes in the UPS have been proposed to enhance polyubiquitination as well. Desonide has shown promising effects in suppressing mHTT expression, but it remains uncertain whether the desonide-mHTT interaction caused the reduction of mHTT-induced toxicity [163]. Other compounds whose mechanism is neither UPS nor autophagy are also summarized in this section despite

Table 1
Types of administration: advantages and limitations.

Types		Advantages	Disadvantages	Refs
Parenchyma delivery	Intrastriatal injection	●Widespread distribution	●Invasive surgical techniques	[56,57,72,74,75, 76,106, 109,140, 141,161,171,172]
CSF delivery	Intrathecal injection	●Straightforward for most patients ●High concentration of therapeutic agent within CNS	●Discomfort and Painful ●Difficult to perform in special populations (obese or abnormal spine anatomy patients) ·Daily administration	[131]
	Intracerebroventricular injection	●Relative consistency of the surgical procedure ●Developed neurosurgery protocols	●Limited data on long-term use ●Parenchyma injury induced by needle	[87,107,19,112, 113,116, 152,162]
Systemic delivery	Intraperitoneal injection	●Easy to master ●Quick procedure ●Low impact of stress on laboratory rodents	●First pass metabolism ●Not suitable for evaluation of properties of a drug	[154]
	Intravenous injection	●Noninvasive	●Limited partitioning to the brain ●The immune barrier and adverse effects ●Pressure on healthcare systems	[46]
	Subcutaneous injection	●Effective and well-tolerated ●Reduced drug delivery-related healthcare costs and resource use	●Difficult to spread larger volumes at a single injection site	[86,120]

the underlying molecular mechanisms remaining unknown. They were identified by a cell-based screen and stimulated selective clearance of mHTT in vitro [164]. Attention should be given to optimizing the small molecules mentioned above are cell permeability, tissue distribution, and pharmacokinetic properties [165].

Current allele-selective intrabodies in HD are mainly single-chain Fv (scFv) intrabodies. They are usually used for reducing aggregation and have a simple molecular structure that is made of H- and L-chain variable (VH and VL) segments held together by a short linker sequence [161,166]. Lecerf et al. and Miller et al. found sFv antibody specific for soluble N-terminal 17 amino acids of HTT but was weakly active against endogenous full-length mHTT and wtHTT, possibly due to epitope inaccessibility [167,168]. Furthermore, scFv antibody raised HD flies survival rate to adulthood from 23 % to 100 % and the mean and maximum lifespan of adult HD flies was significantly prolonged [169]. To improve the neuroprotective effect, scFv was transferred with mouse orthenine decarboxylase (mODC)-fused C-terminal PEST motif which was often labeled for proteasomal degradation. The complex prevented aggregation and reduced HD exon 1 fragment toxicity at low doses and inhibited degradation of scFv themselves, implying the additional optimization of scFv would be beneficial for future use in clinical applications [170]. Other studies identified that MW7 and Happ1 antibodies recognized the polyP and P-rich domain of HTT respectively and reduced mHDx-1 aggregation and toxicity in HD cell culture and increased turnover of mutant but not wtHDx-1 [171]. In five HD mouse models (lentiviral, R6/2, N171–82Q, YAC128, and BACHD models), the Happ1 treatment provided highly beneficial effects across various assays assessing motor and cognitive deficits, while exhibiting a very low probability of off-target effects. But it only resulted in improvements in body weight and an extension of lifespan in N171–82Q mice. These results support the idea that intrabodies directed at the selective reduction of mHTT aggregation represent an effective strategy for HD treatment [172]. Wang et al. developed scFv EM48, which recognized an epitope C-terminal to the P-rich region of HTT and preferentially bound with mHTT and increased turnover of mHDx-1 in vitro. When expressed in the striatum of HD mice via AAV infection, the intrabody reduced neuropil aggregation formation and improved some aspects of neuropathology as well as motor performance [173]. A part of the above-mentioned intrabodies are too large to cross the BBB and cytosolic expression of intrabodies did not always result in functional antibodies as they tend to misfold in the cytosol, the efficiency to remove mHTT inside neuronal cells and physicochemical parameters of intrabodies also need be improved [174].

6. Discussion

Until now, it has been 30 years since the HD mutant gene was identified in 1993. Allele-selective mHTT-lowering strategies have revealed new dimensions of HD treatment. DNA-based approaches are the most proximal therapeutic target in the mHTT pathway making them a powerful disease-modifying strategy for HD. RNA-based therapeutics are also prospective and earlier employed in HD human trials. In recent years targeted mHTT degradation, as a therapeutic modality, has seen substantial progress in elevating protein degradation rates and preventing protein aggregation. Although these approaches expanded the scope of the treatment of HD, there are exist issues about translating therapeutic HD paradigms into the clinic. Overall, DNA-based therapeutics can cause immunogenicity by bacterial proteins or of new edited genes and modify the germline with irreversible genome changes and ethical concerns [175]. RNA therapeutics face several challenges, including issues related to specificity, stability, toxicity, and immune activation. Lacking the principle of designing, the low BBB penetration, and the insufficient understanding of autophagy and the UPS limit the development of small molecules targeting protein degradation. Besides, all allele-selective mHTT-lowering strategies share the challenges of administration, animal models, drug delivery and bring additional liabilities of their own. Table 1 provides a comparative analysis of the advantages and limitations associated with each type of administration.

The latest study used pigs to test the efficacy and safety of therapies, thereby narrowing the anatomical and metabolic regulation gaps between animal models and humans [46]. Nevertheless, mice remain the most commonly used animal species. There are two broad categories: transgenic models and knock-in (KI) models, which allow researchers to evaluate the efficacy of therapeutic compounds during the preclinical development stage of HD. Table 2 sums up CAG repeats and main features including symptoms and application of each HD mouse model. There are generally some points to be mentioned and noticed. (a) Most HD mouse models carry 110–250 CAG repeats and the establishment of mice within the typical human range is required [176]. (b) The discrepancies in body weight changes, onset of HD phenotypes, and progression exist in different HD mouse models. R6/2 demonstrated weight loss, whereas full-length murine HD models, such as YAC128, BACHD, Hu97/18, and Hu128/21 mice gained weight [119]. The R6 mice exhibit rapid and severe HD pathology that is widespread. Contrarily, most of the KI and full-length rodent models have a less progressive phenotype and region-specific neuropathology. (c) Current murine models are insufficient to accurately represent human HD. The R6 mice harbor two normal mouse copies and one mutated human copy of the gene, which are different from humans. N171–82Q model expresses

Table 2
Summary of HD mouse models.

Induction mechanism	Mouse model	CAG repeat size	Main features	Refs
Transgenic-induced	R6/2	114	●Aggressive and rapidly progressive behaviour phenotypes	[56,57,112,120,161,171,172]
	R6/1	116	(Compared with R6/2) ●Simlar behaviour changes ●Postponed onset by several weeks ●Slower progression of the severity of the symptoms	[167]
	BACHD	97	●Contain large segment genomic DNA ●A mixed CAG/CAA repeat ●mHTT expression: 150–200 % of the endogenous murine HD homolog ●Fewer mHTT aggregates (Compared with YAC) ●Display relatively robust motor deficits and brain atrophy	[86,87,106,171]
	YAC	128	●Contain large segment genomic DNA ●A relatively pure CAG repeat (Compared with BACHD) ●mHTT expression: 150–200 % of the endogenous murine HD homolog ●Confer Milder deficits (Compared with BACHD)	[76,106,171]
	Hu128/21 (YAC128/BAC21)	/	●Suitable for evaluating the drugs targeting SNPs or different CAG lengths	[74,75]
	Hu97/18 (BACHD/YAC18)		●Do not fit in assessing CAG-targeting approaches	[107,109,113]
	N171–82Q	82	●Later onset (Compared with R6/2) ●Lesser Motor symptoms (tremors, hypokinesia, and clasping of the limbs)	[116,171,172]
Knock-in induced	ZQ175	ca. 190	●Not remarkable behavioral dysfunction	[57,140,141]
	Hdh-Q140	140		[152,154,162]
	HTT Q50	50		[57]

mHTT only in neurons but not glia, which ignores the principal factor contributing to neuronal excitotoxicity. Few studies report KI models that have substantive neuropathology in the KI mouse strain. And most literature predominantly describes the presence of cellular inclusions and striatal astrogliosis, with limited evidence of significant neuronal loss [177]. Given that many differences exist in human and animal models, a further comprehensive understanding of HD rodent models would be necessary in the future.

Crossing the BBB and disseminating it to lesions in a broad brain region is technically challenging. The current research on HD primarily focuses on viral vectors for delivering vehicles. The commonly used viral vectors are AAVs and lentiviruses (LVs). The AAV vectors are usually preferred for in vivo gene therapy due to several advantages, including low innate immune response, the ability to be permanently effective through a single administration, and robust transduction efficiency. The limitations of AAVs are the insert size, delay in transgene expression, targeted tissue specificity, and high cost, stressing the necessity of further optimization methods. LVs also show minimal inflammatory response and have high transduction efficiency like AAVs. They can offer several potentially unique advantages over AAVs. Neutralizing antibodies are rarely generated against LVs, meaning they can stably integrate into the genome of host cells in the long term. But an important safety concern for LVs is the risk of insertional mutagenesis, which is opposite to AAVs in that the transgene is mainly episomally expressed and superiorly spread in vivo because of the relatively small size of the viral particles [178,179].

7. Conclusions

Since HD is linked to aberrant accumulation of aggression-prone mHTT and lowering mHTT would ameliorate and even reverse pathogenic effects, the approaches targeting *mutant HTT* DNA, RNA, and itself have been developed for disease modification. Given that wild-type *HTT* is crucial for brain development and neurogenesis during development and adulthood and the loss of wtHTT have potential safety problems, allele-selective methods offer a more suitable choice for HD treatment. All of them mainly contain SNP- and CAG repeat-targeting approaches. Nowadays the clinical trials about WVE-003 are ongoing at present, indicating promising prospects for therapies that selectively target and reduce mHTT levels. Improved delivery vectors and administrations, choices of animal models, pharmacological properties, safety and durability, deep researches on the mechanism of off-target effects, and intervention earlier in the time course of HD pathology will help promote the design of drug candidates.

Ethical approval and consent to participate

This research was a review, and no participants took part, so there was no need for it.

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CRediT authorship contribution statement

Jia-Yuan Yao: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Ting Liu:** Writing – review & editing, Investigation. **Xin-ru Hu:** Writing – review & editing, Investigation. **Hui Sheng:** Writing – review & editing, Investigation. **Zi-hao Chen:** Writing – review & editing, Investigation. **Hai-yang Zhao:** Writing – review & editing, Investigation. **Xiao-jia Li:** Writing – review & editing, Supervision, Conceptualization. **Yang Wang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization. **Liang Hao:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data availability is not applicable to this article as no new data were created or analyzed in this study.

References

[1] G.P. Bates, R. Dorsey, J.F. Gusella, M.R. Hayden, C. Kay, B.R. Leavitt, M. Nance, C.A. Ross, R.I. Scahill, R. Wetzel, E.J. Wild, S.J. Tabrizi, Huntington disease, Nat. Rev. Dis. Prim. 1 (2015) 15005, https://doi.org/10.1038/nrdp.2015.5.

[2] E. Cubo, S.I. Martinez-Horta, F.S. Santalo, A.M. Descalls, S. Calvo, C. Gil-Polo, I. Muñoz, K. Llano, N. Mariscal, D. Diaz, A. Gutierrez, L. Aguado, M.A. Ramos-Arroyo, Clinical manifestations of homozygote allele carriers in Huntington disease, Neurology 92 (18) (2019) e2101–e2108, https://doi.org/10.1212/WNL.0000000000007147.

[3] M.A. Nance, Huntington disease: clinical, genetic, and social aspects, J. Geriatr. Psychiatry Neurol. 11 (2) (1998) 61–70, https://doi.org/10.1177/089198879801100204.

[4] F.O. Walker, Huntington's disease, Lancet 369 (9557) (2007) 218–228, https://doi.org/10.i1055/s-2007-971176.

[5] S.E. Andrew, Y.P. Goldberg, B. Kremer, H. Telenius, J. Theilmann, S. Adam, E. Starr, F. Squitieri, B. Lin, M.A. Kalchman, The relationship between trinucleotide (CAG) repeat length and clinical features of Huntington's disease, Nat. Genet. 4 (1993) 398–403, https://doi.org/10.1038/ng0893-398.

[6] J.M. Lee, E.M. Ramos, J.H. Lee, T. Gillis, J.S. Mysore, M.R. Hayden, S.C. Warby, P. Morrison, M. Nance, C.A. Ross, R.L. Margolis, CAG repeat expansion in Huntington disease determines age at onset in a fully dominant fashion, Neurology 78 (10) (2012) 690–695, https://doi.org/10.1212/WNL.0b013e318249f683.

[7] J.M. Lee, K. Correia, J. Loupe, K.H. Kim, D. Barker, E.P. Hong, M.J. Chao, J. D. Long, D. Lucente, J.P.G. Vonsattel, R.M. Pinto, CAG repeat not polyglutamine length determines timing of Huntington's disease onset, Cell 178 (4) (2019) 887–900, https://doi.org/10.1016/j.cell.2019.06.036.

[8] S.J. Tabrizi, R. Ghosh, B.R. Leavitt, Huntingtin lowering strategies for disease modification in huntington's disease, Neuron 101 (5) (2019) 801–819, https://doi.org/10.1016/j.neuron.2019.01.039.

[9] A.S. Taran, L.D. Shuvalova, M.A. Lagarkova, I.B. Alieva, Huntington's disease-an outlook on the interplay of the HTT protein, microtubules and actin cytoskeletal components, Cells 9 (2020) 1514, https://doi.org/10.3390/cells9061514.

[10] E. Fields, E. Vaughan, D. Tripu, I. Lim, K. Shrout, J. Conway, N. Salib, Y. Lee, A. Dhamsania, M. Jacobsen, A. Woo, H. Xue, K. Cao, Gene targeting techniques for Huntington's disease, Ageing Res. Rev. 70 (2021) 101385, https://doi.org/10.1016/j.arr.2021.101385.

[11] H. Bashir, Emerging therapies in Huntington's disease, Expert Rev. Neurother. 19 (10) (2019) 983–995, https://doi.org/10.1080/14737175.2019.1631161.

[12] B.R. Leavitt, H.B. Kordasiewicz, S.A. Schobel, Huntingtin-lowering therapies for huntington disease: a review of the evidence of potential benefits and risks, JAMA Neurol. 77 (2020) 764–772, https://doi.org/10.1001/jamaneurol.2020.0299.

[13] C. Estevez-Fraga, F.B. Rodrigues, S.J. Tabrizi, E.J. Wild, Huntington's disease clinical trials corner: april 2022, J. Huntingt. Dis. 11 (2022) 105–118, https://doi.org/10.3233/JHD-229002.

[14] E.F. Stimming, V. Sung, C. Testa, S. Kostyk, C.A. Ross, A. Samii, M.D. Geschwind, D. Hall, P. Dayalu, R. Lonser, B. Elder, J05 Interim results from cohort 1 of the double-blind, dose-escalation phase I/II clinical trial of amt-130 (HD-genetrx-1) for early-stage huntington's disease (HD), J. Neurol. Neurosur. Ps 93 (Suppl 1) (2022), https://doi.org/10.1136/jnnp-2022-ehdn.255.

[15] S. Upadhayay, S. Jamwal, P. Kumar, Animal models of Huntington's disease and their applicability to novel drug discovery and development, Expert. Opin. Drug. Discov. 18 (5) (2023) 527–538, https://doi.org/10.1080/17460441.2023.2201493.

[16] J.P. Liu, S.O. Zeitlin, Is huntingtin dispensable in the adult brain, J. Huntingt. Dis. 6 (1) (2017) 1–17, https://doi.org/10.3233/JHD-170235.

[17] R.M. Bragg, E.W. Mathews, A. Grindeland, J.P. Cantle, D. Howland, T. Vogt, J. B. Carroll, Global huntingtin knockout in adult mice leads to fatal neurodegeneration that spares the pancreas, Life. Sci. Alliance 7 (9) (2024) e202402571, https://doi.org/10.26508/lsa.202402571.

[18] M. Kaliszewski, A.B. Knott, E. Bossy-Wetzel, Primary cilia and autophagic dysfunction in Huntington's disease, Cell. Death. Differ. 22 (2015) 1413–1424, https://doi.org/10.1038/cdd.2015.80.

[19] F. Saudou, S. Humbert, 2016, , Biol. Hunt. Neuron 89910–926, 10.1016/j.jconrel.2016.09.011.

[20] M. Jimenez-Sanchez, F. Licitra, B.R. Underwood, D.C. Rubinsztein, Huntington's disease: mechanisms of pathogenesis and therapeutic strategies, Cold. Spring Harb. Perspect. Med. 7 (2017) a024240, https://doi.org/10.1101/cshperspect.a024240.

[21] B.P. Imbimbo, V. Triaca, C. Imbimbo, R. Nisticò, Investigational treatments for neurodegenerative diseases caused by inheritance of gene mutations: lessons from recent clinical trials, Neural Regen. Res. 18 (2023) 1679–1683, https://doi.org/10.4103/1673-5374.363185.

[22] J.C. Barron, F. Nafar, M.P. Parsons, Huntingtin is essential for synaptic plasticity in the adult hippocampus, bioRxiv 10 (2022), https://doi.org/10.1101/2022.10.05.510980.

[23] I. Dragatsis, M.S. Levine, S. Zeitlin, Inactivation of Hdh in the brain and testis results in progressive neurodegeneration and sterility in mice, Nat. Genet. 26 (2000) 300–306, https://doi.org/10.1038/81593.

[24] W. Auerbach, M.S. Hurlbert, P. Hilditch-Maguire, Y.Z. Wadghiri, V.C. Wheeler, S. I. Cohen, A.L. Joyner, M.E. MacDonald, D.H. Turnbull, The HD mutation causes progressive lethal neurological disease in mice expressing reduced levels of huntingtin, Hum. Mol. Genet. 10 (2001) 2515–2523, https://doi.org/10.1093/hmg/10.22.2515.

[25] S. Zhang, M.B. Feany, S. Saraswati, J.T. Littleton, N. Perrimon, Inactivation of Drosophila Huntingtin affects long-term adult functioning and the pathogenesis of a Huntington's disease model, Dis. Model. Mech. 2 (2009) 247–266, https://doi.org/10.1038/s41591-019-0478-3.

[26] E.E. Arteaga-Bracho, M. Gulinello, M.L. Winchester, N. Pichamoorthy, J. R. Petronglo, A.D. Zambrano, J. Inocencio, C.D. De Jesus, J.O. Louie, S. Gokhan, M.F. Mehler, A.E. Molero, Postnatal and adult consequences of loss of huntingtin during development: implications for Huntington's disease, Neurobiol. Dis. 96 (2016) 144–155, https://doi.org/10.1016/j.nbd.2016.09.006.

[27] M.F. Mehler, J.R. Petronglo, E.E. Arteaga-Bracho, M.E. Gulinello, M. L. Winchester, N. Pichamoorthy, S.K. Young, C.D. DeJesus, H. Ishtiaq, S. Gokhan, A.E. Molero, Loss-of-Huntingtin in medial and lateral ganglionic lineages differentially disrupts regional interneuron and projection neuron subtypes and promotes Huntington's disease-associated behavioral, cellular, and pathological hallmarks, J. Neurosci. 39 (2019) 1892–1909, https://doi.org/10.1523/JNEUROSCI.2443-18.2018.

[28] B.Y. Braz, D. Wennagel, L. Ratié, D.A.R. de Souza, J.C. Deloulme, E.L. Barbier, A. Buisson, F. Lanté, S. Humbert, Treating early postnatal circuit defect delays Huntington's disease onset and pathology in mice, Science 377 (2022) eabq5011, https://doi.org/10.1126/science.abq5011.

[29] Y.M. Sun, Y.B. Zhang, Z.Y. Wu, Huntington's disease: relationship between phenotype and genotype, Mol. Neurobiol. 54 (1) (2017) 342–348, https://doi.org/10.1007/s12035-015-9662-8.

[30] C. Kay, J.A. Collins, N.H. Skotte, A.L. Southwell, S.C. Warby, N.S. Caron, C. N. Doty, B. Nguyen, A. Griguoli, C.J. Ross, F. Squitieri, Huntingtin haplotypes provide prioritized target panels for allele-specific silencing in Huntington disease patients of European ancestry, Mol. Ther. 23 (11) (2015) 1759–1771, https://doi.org/10.1038/mt.2015.128.

[31] C. Kay, J.A. Collins, N.S. Caron, L.A. Agostinho, H. Findlay-Black, L. Casal, D. Sumathipala, V.H.W. Dissanayake, M. Cornejo-Olivas, F. Baine, A. Krause, J. L. Greenberg, C.L.A. Paiva, F. Squitieri, M.R. Hayden, A comprehensive haplotype-targeting strategy for allele-specific HTT suppression in huntington disease, Am. J. Hum. Genet. 105 (6) (2019) 1112–1125, https://doi.org/10.1016/j.ajhg.2019.10.011.

[32] B.R. Leavitt, Current clinical trials of new therapeutic agents for Huntington's disease. In Huntington's Disease, Academic Press, 2024, pp. 571–589, https://doi.org/10.1016/B978-0-323-95672-7.00015-7.

[33] J. Wu, B. Tang, Y. Tang, Allele-specific genome targeting in the development of precision medicine, Theranostics 10 (7) (2020) 3118–3137, https://doi.org/10.7150/thno.43298.

[34] S.H. Sternberg, S. Redding, M. Jinek, E.C. Greene, J.A. Doudna, DNA interrogation by the CRISPR RNA-guided endonuclease Cas9, Nature 507 (2014) 62–67, https://doi.org/10.1038/nature13011.

[35] B.P. Kleinstiver, M.S. Prew, S.Q. Tsai, V.V. Topkar, N.T. Nguyen, Z. Zheng, A. P. Gonzales, Z. Li, R.T. Peterson, J.R. Yeh, M.J. Aryee, J.K. Joung, Engineered CRISPR-Cas9 nucleases with altered PAM specificities, Nature 523 (7561) (2015) 481–485, https://doi.org/10.1038/nature14592.

[36] J.W. Shin, K.H. Kim, M.J. Chao, R.S. Atwal, T. Gillis, M.E. MacDonald, J. F. Gusella, J.M. Lee, Permanent inactivation of Huntington's disease mutation by personalized allele-specific CRISPR/Cas9, Hum. Mol. Genet. 25 (2016) 4566–4576, https://doi.org/10.1093/hmg/ddw286.

[37] A.M. Monteys, S.A. Ebanks, M.S. Keiser, B.L. Davidson, CRISPR/Cas9 editing of the mutant huntingtin allele in vitro and in vivo, Mol. Ther. 25 (2017) 12–23, https://doi.org/10.1016/j.ymthe.2016.11.010.

[38] H. Malkki, Selective deactivation of huntington disease mutant allele by crispr–cas9 gene editing, Nat. Rev. Neurol. 12 (11) (2016) 614–615, https://doi.org/10.1038/nrneurol.2016.151.

[39] S. Rosignoli, E. Lustrino, A. Conci, A. Fabrizi, S. Rinaldo, M.C. Latella, E. Enzo, G. Prosseda, L. De Rosa, M. De Luca, A. Paiardini, AlPaCas: allele-specific CRISPR gene editing through a protospacer-adjacent-motif (PAM) approach, Nucleic Acids Res 52 (W1) (2024) W29–W38, https://doi.org/10.1093/nar/gkae419.

[40] S.R. Oikemus, E.L. Pfister, E. Sapp, K.O. Chase, L.A. Kennington, E. Hudgens, R. Miller, L.J. Zhu, A. Chaudhary, E.O. Mick, M. Sena-Esteves, S.A. Wolfe, M. DiFiglia, N. Aronin, M.H. Brodsky, Allele-specific knockdown of mutant huntingtin protein via editing at coding region single nucleotide polymorphism

heterozygosities, Hum. Gene. Ther. 33 (1-2) (2022) 25–36, https://doi.org/10.1089/hum.2020.323.
[41] J.W. Shin, E.P. Hong, S.S. Park, D.E. Choi, I.S. Seong, M.N. Whittaker, B. P. Kleinstiver, R.Z. Chen, J.M. Lee, Allele-specific silencing of the gain-of-function mutation in Huntington's disease using CRISPR/Cas9, Jci. Insight 7 (19) (2022) e141042, https://doi.org/10.1172/jci.insight.141042.
[42] J.W. Shin, E.P. Hong, S.S. Park, D.E. Choi, S. Zeng, R.Z. Chen, J.M. Lee, PAM-altering SNP-based allele-specific CRISPR-Cas9 therapeutic strategies for Huntington's disease, Mol. Ther. Methods Clin. Dev. 26 (2022) 547–561, https://doi.org/10.1016/j.omtm.2022.08.005.
[43] M. Dabrowska, W. Juzwa, W.J. Krzyzosiak, M. Olejniczak, Precise Excision of the CAG Tract from the Huntingtin Gene by Cas9 Nickases, Front. Neurosci. 12 (2018) 75, https:// doi.org/10.3389/fnins.2018.00075.
[44] A. Murillo, M. Larin, E.L. Randall, A. Taylor, M. Lelos, V. Dion, I05 CRISPR-Cas9 nickase-mediated gene editing to treat Huntington's disease, J. Neurol. Neurosurg. Psychiatry (2022), https://doi.org/10.1136/jnnp-2022-ehdn.231.
[45] X. Xu, Y. Tay, B. Sim, S.I. Yoon, Y. Huang, J. Ooi, K.H. Utami, A. Ziaei, B. Ng, C. Radulescu, D. Low, A.Y.J. Ng, M. Loh, B. Venkatesh, F. Ginhoux, G. J. Augustine, M.A. Pouladi, Reversal of phenotypic abnormalities by CRISPR/Cas9-mediated gene correction in Huntington disease patient-derived induced pluripotent stem cells, Stem. Cell. Rep. 8 (2017) 619–633, https://doi.org/10.1016/j.stemcr.2017.01.022.
[46] S. Yan, X. Zheng, Y. Lin, C. Li, Z. Liu, J. Li, Z. Tu, Y. Zhao, C. Huang, Y. Chen, J. Li, X. Song, B. Han, W. Wang, W. Liang, L. Lai, X.J. Li, S. Li, Cas9-mediated replacement of expanded CAG repeats in a pig model of Huntington's disease, Nat. Biomed. Eng. 7 (2023) 629–646, https://doi.org/10.1038/s41551-023-01007-3.
[47] C.A. de Solis, A. Ho, R. Holehonnur, J.E. Ploski, The development of a viral mediated CRISPR/Cas9 system with doxycycline dependent gRNA expression for inducible in vitro and in vivo genome editing, Front. Mol. Neurosci. 9 (2016) 70, https://doi.org/10.3389/fnmol.2016.00070.
[48] M. Luo, L.K.C. Lee, B. Peng, C.H.J. Choi, W.Y. Tong, N.H. Voelcker, Delivering the promise of gene therapy with nanomedicines in treating central nervous system diseases, Adv. Sci. 9 (2022) e2201740, https://doi.org/10.1002/advs.202201740.
[49] D. Wang, P.W.L. Tai, G. Gao, Adeno-associated virus vector as a platform for gene therapy delivery, Nat. Rev. Drug. Discov. 18 (2019) 358–378, https://doi.org/10.1038/s41573-019-0012-9.
[50] N. Deglon, Gabriel Vachey, M. Rey, A. Perrier, I07Allele specific gene editing for huntington's disease mediated by the KAMICAS9 self-inactivating CRISPR/CAS9 system, J. Neurol. Neurosurg. Psychiatry (2018), https://doi.org/10.1136/jnnp-2018-EHDN.243.
[51] M. Eisenstein, CRISPR takes on Huntington's disease, Nature 557 (2018) S42–S43, https://doi.org/10.1038/d41586-018-05177-y.
[52] H.A. Han, J.K.S. Pang, B.S. Soh, Mitigating off-target effects in CRISPR/Cas9-mediated in vivo gene editing, J. Mol. Med. 98 (5) (2020) 615–632, https://doi.org/10.1186/10.1007/s00109-020-01893-z.
[53] C. Li, Y. Lin, Y. Chen, X. Song, X. Zheng, J. Li, J. He, X. Chen, C. Huang, W. Wang, J. Wu, J. Wu, J. Gao, Z. Tu, X.J. Li, S. Yan, S. Li, A specific mini-intrabody mediates lysosome degradation of mutant huntingtin, Adv. Sci. 10 (2023) e2301120, https://doi.org/10.1002/advs.202301120.
[54] J. Yao, J. Huang, T. Hai, X. Wang, G. Qin, H. Zhang, R. Wu, C. Cao, J.J. Xi, Z. Yuan, J. Zhao, Efficient bi-allelic gene knockout and site-specific knock-in mediated by TALENs in pigs, Sci. Rep. 4 (2014) 6926, https://doi.org/10.1038/srep06926.
[55] Y. Mineta, T. Okamoto, K. Takenaka, N. Doi, Y. Aoyama, T. Sera, Enhanced cleavage of double-stranded DNA by artificial zinc-finger nuclease sandwiched between two zinc-finger proteins, Biochemistry 47 (2008) 12257–12259, https://doi.org/10.1021/bi801800k.
[56] M. Garriga-Canut, C. Agustín-Pavón, F. Herrmann, A. Sánchez, M. Dierssen, C. Fillat, M. Isalan, Synthetic zinc finger repressors reduce mutant huntingtin expression in the brain of R6/2 mice, Proc. Natl. Acad. Sci. Usa. 109 (2012) E3136–E3145, https://doi.org/10.1073/pnas.1206506109.
[57] B. Zeitler, S. Froelich, K. Marlen, D.A. Shivak, Q. Yu, D. Li, J.R. Pearl, J.C. Miller, L. Zhang, D.E. Paschon, S.J. Hinkley, I. Ankoudinova, S. Lam, D. Guschin, L. Kopan, J.M. Cherone, H.B. Nguyen, G. Qiao, Y. Ataei, M.C. Mendel, R. Amora, R. Surosky, J. Laganiere, B.J. Vu, A. Narayanan, Y. Sedaghat, K. Tillack, C. Thiede, A. Gärtner, S. Kwak, J. Bard, L. Mrzljak, L. Park, T. Heikkinen, K. K. Lehtimäki, M.M. Svedberg, J. Häggkvist, L. Tari, M. Tóth, A. Varrone, C. Halldin, A.E. Kudwa, S. Ramboz, M. Day, J. Kondapalli, D.J. Surmeier, F. D. Urnov, P.D. Gregory, E.J. Rebar, I. Muñoz-Sanjuán, H.S. Zhang, Allele-selective transcriptional repression of mutant HTT for the treatment of Huntington's disease, Nat. Med. 25 (2019) 1131–1142, https://doi.org/10.1038/s41591-019-0478-3.
[58] D. Kim, K. Luk, S.A. Wolfe, J.S. Kim, Evaluating and enhancing target specificity of gene-editing nucleases and deaminases, Annu. Rev. Biochem. 88 (1) (2019) 191–220, https://doi.org/10.1146/annurev-biochem-013118-111730.
[59] J. Li, J.J. Røise, M. He, R. Das, N. Murthy, Non-viral strategies for delivering genome editing enzymes, Adv. Drug. Deliv. Rev. 168 (2021) 99–117, https://doi.org/10.1016/j.addr.2020.09.004.
[60] A. Shahryari, I. Burtscher, Z. Nazari, H. Lickert, Engineering gene therapy: advances and barriers, Adv. Ther. 4 (2021) 2100040, https://doi.org/10.1002/adtp.202100040.
[61] G.F. Richard, D. Viterbo, V. Khanna, V. Mosbach, L. Castelain, B. Dujon, Highly specific contractions of a single CAG/CTG trinucleotide repeat by TALEN in yeast, PloS. One 9 (2014) e95611, https://doi.org/10.1371/journal.pone.0095611.
[62] K.D. Fink, P. Deng, J. Gutierrez, J.S. Anderson, A. Torrest, A. Komarla, S. Kalomoiris, W. Cary, J.D. Anderson, W. Gruenloh, A. Duffy, T. Tempkin, G. Annett, V. Wheelock, D.J. Segal, J.A. Nolta, Allele-specific reduction of the mutant huntingtin allele using transcription activator-like effectors in human huntington's disease fibroblasts, Cell. Transplant. 25 (2016) 677–686, https://doi.org/10.3727/096368916X690863.
[63] A. Jurcau, C.M. Jurcau, Mitochondria in Huntington's disease: implications in pathogenesis and mitochondrial-targeted therapeutic strategies, NRR 18 (7) (2023) 1472–1477, https://doi.org/10.4103/1673-5374.360289.
[64] M. Nakamori, G.B. Panigrahi, S. Lanni, T. Gall-Duncan, H. Hayakawa, H. Tanaka, J. Luo, T. Otabe, J. Li, A. Sakata, M.C. Caron, N. Joshi, T. Prasolava, K. Chiang, J. Y. Masson, M.S. Wold, X. Wang, M.Y.W.T. Lee, J. Huddleston, K.M. Munson, S. Davidson, M. Layeghifard, L.M. Edward, R. Gallon, M. Santibanez-Koref, A. Murata, M.P. Takahashi, E.E. Eichler, A. Shlien, K. Nakatani, H. Mochizuki, C. E. Pearson, A slipped-CAG DNA-binding small molecule induces trinucleotide-repeat contractions in vivo, Nat. Genet. 52 (2020) 146–159, https://doi.org/10.1038/s41588-019-0575-8.
[65] J. Li, A. Sakata, H. He, L.P. Bai, A. Murata, C. Dohno, K. Nakatani, Naphthyridine-benzoazaquinolone: evaluation of a tricyclic system for the binding to (CAG)n repeat DNA and RNA, Chem. Asian J. 11 (2016) 1971–1981, https://doi.org/10.1002/asia.201600527.
[66] T.M. Geel, M.H.J. Ruiters, R.H. Cool, L. Halby, D.C. Voshart, L. Andrade Ruiz, K. E. Niezen-Koning, P.B. Arimondo, M.G. Rots, The past and presence of gene targeting: from chemicals and DNA via proteins to RNA, Philos. Trans. R. Soc. Lond. B. Biol. Sci. 373 (2018) 20170077, https://doi.org/10.1098/rstb.2017.0077.
[67] G.M. Traber, A.M. Yu, RNAi-based therapeutics and novel RNA bioengineering Technologies, J. Pharmacol. Exp. Ther. 384 (2023) 133–154, https://doi.org/10.1124/jpet.122.001234.
[68] R.C.C. Ryther, A.S. Flynt, J. 3 Phillips, J.G. Patton, siRNA therapeutics: big potential from small RNAs, Gene. Ther. 12 (2005) 5–11, https://doi.org/10.1038/sj.gt.3302356.
[69] R.W. Carthew, E.J. Sontheimer, Origins and mechanisms of miRNAs and siRNAs, Cell 136 (2009) 642–655, https://doi.org/10.1016/j.cell.2009.01.035.
[70] J.K. Lam, M.Y. Chow, Y. Zhang, S.W. Leung, siRNA versus miRNA as therapeutics for gene silencing, Mol. Ther. Nucleic Acids 4 (2015) e252, https://doi.org/10.1038/mtna.2015.23.
[71] J. Miniarikova, M.M. Evers, P. Konstantinova, Translation of MicroRNA-based huntingtin-lowering therapies from preclinical studies to the clinic, Mol. Ther. 26 (2018) 947–962, https://doi.org/10.1016/j.ymthe.2018.02.002.
[72] A.M. Monteys, M.J. Wilson, R.L. Boudreau, R.M. Spengler, B.L. Davidson, Artificial miRNAs targeting mutant huntingtin show preferential silencing in vitro and in vivo. molecular therapy, Mol. Ther. Nucleic Acids 4 (2015) e234, https://doi.org/10.1038/mtna.2015.7.
[73] M.M. Maxwell, RNAi applications in therapy development for neurodegenerative disease, Curr. Pharm. Des. 15 (2009) 3977–3991, https://doi.org/10.2174/138161209789649295.
[74] J. Miniarikova, I. Zanella, A. Huseinovic, T. van der Zon, E. Hanemaaijer, R. Martier, A. Koornneef, A.L. Southwell, M.R. Hayden, S.J. van Deventer, H. Petry, P. Konstantinova, Design, characterization, and lead selection of therapeutic miRNAs targeting huntingtin for development of gene therapy for huntington's disease, Mol. Ther. Nucleic Acids 5 (2016) e297, https://doi.org/10.1038/mtna.2016.7.
[75] J. Miniarikova, V. Zimmer, R. Martier, C.C. Brouwers, C. Pythoud, K. Richetin, M. Rey, J. Lubelski, M.M. Evers, S.J. van Deventer, H. Petry, N. Déglon, P. Konstantinova, AAV5-miHTT gene therapy demonstrates suppression of mutant huntingtin aggregation and neuronal dysfunction in a rat model of Huntington's disease, Gene. Ther. 24 (2017) 630–639, https://doi.org/10.1038/gt.2017.71.
[76] A. Kotowska-Zimmer, L. Przybyl, M. Pewinska, J. Suszynska-Zajczyk, D. Wronka, M. Figiel, M. Olejniczak, A CAG repeat-targeting artificial miRNA lowers the mutant huntingtin level in the YAC128 model of Huntington's disease, Mol. Ther. Nucleic Acids 28 (2022) 702–715, https://doi.org/10.1016/j.omtn.2022.04.031.
[77] A. Ciesiolka, A. Stroynowska-Czerwinska, P. Joachimiak, A. Ciolak, E. Kozlowska, M. Michalak, M. Dabrowska, M. Olejniczak, K.D. Raczynska, D. Zielinska, M. Wozna-Wysocka, W.J. Krzyzosiak, A. Fiszer, Artificial miRNAs targeting CAG repeat expansion in ORFs cause rapid deadenylation and translation inhibition of mutant transcripts, Cell. Mol. Life. Sci. 78 (2021) 1577–1596, https://doi.org/10.1007/s00018-020-03596-7.
[78] A.L. Jackson, P.S. Linsley, Recognizing and avoiding siRNA off-target effects for target identification and therapeutic application, Nat. Rev. Drug. Discov. 9 (2010) 57–67, https://doi.org/10.1038/nrd3010.
[79] B.L. Davidson, H.L. Paulson, Molecular medicine for the brain: silencing of disease genes with RNA interference, Lancet Neurol. 3 (2004) 145–149, https://doi.org/10.1016/S1474-4422(04)00678-7.
[80] B.M. Godinho, M. Malhotra, C.M. O'Driscoll, J.F. Cryan, Delivering a disease-modifying treatment for Huntington's disease, Drug. Discov. Today 20 (2015) 50–64, https://doi.org/10.1016/j.drudis.2014.09.011.
[81] M.C.P. Mendonça, A. Kont, M.R. Aburto, J.F. Cryan, C.M. O'Driscoll, Advances in the design of (Nano)formulations for delivery of antisense oligonucleotides and small interfering RNA: focus on the central nervous system, Mol. Pharm. 18 (2021) 1491–1506, https://doi.org/10.1021/acs.molpharmaceut.0c01238.
[82] A. Mas-Monteys, S.Q. Harper, B.L. Gilmore, P.D. Staber, C. Schaffer, B. Polisky, C. Vargeese, B.L. Davidson, 711. allele-specific silencing of mutant huntingtin for huntington's disease therapy, Mol. Ther. 13 (2006) S274–S275, https://doi.org/10.1016/j.ymthe.2006.08.790.

[83] P.H. van Bilsen, L. Jaspers, M.S. Lombardi, J.C. Odekerken, E.N. Burright, W. F. Kaemmerer, Identification and allele-specific silencing of the mutant huntingtin allele in Huntington's disease patient-derived fibroblasts, Hum. Gene. Ther. 19 (2008) 710–719, https://doi.org/10.1089/hum.2007.116.
[84] J. Hu, M. Matsui, K.T. Gagnon, J.C. Schwartz, S. Gabillet, K. Arar, J. Wu, I. Bezprozvanny, D.R. Corey, Allele-specific silencing of mutant huntingtin and ataxin-3 genes by targeting expanded CAG repeats in mRNAs, Nat. Biotechnol. 27 (2009) 478–484, https://doi.org/10.1038/nbt.1539.
[85] Y. Zhang, J. Engelman, R.M. Friedlander, Allele-specific silencing of mutant Huntington's disease gene, J. Neurochem. 108 (2009) 82–90, https://doi.org/10.1111/j.1471-4159.2008.05734.x.
[86] K. Yamada, S. Hildebrand, S.M. Davis, R. Miller, F. Conroy, E. Sapp, J. Caiazzi, J. F. Alterman, L. Roux, D. Echeverria, M.R. Hassler, E.L. Pfister, M. DiFiglia, N. Aronin, A. Khvorova, Structurally constrained phosphonate internucleotide linkage impacts oligonucleotide-enzyme interaction, and modulates siRNA activity and allele specificity, Nucleic Acids Res. 49 (2021) 12069–12088, https://doi.org/10.1093/nar/gkab1126.
[87] F. Conroy, R. Miller, J.F. Alterman, M.R. Hassler, D. Echeverria, B.M.D. C. Godinho, E.G. Knox, E. Sapp, J. Sousa, K. Yamada, F. Mahmood, A. Boudi, K. Kegel-Gleason, M. DiFiglia, N. Aronin, A. Khvorova, E.L. Pfister, Chemical engineering of therapeutic siRNAs for allele-specific gene silencing in Huntington's disease models, Nat. Commun. 13 (2022) 5802, https://doi.org/10.1038/s41467-022-33061-x.
[88] J. Hu, J. Liu, D.R. Corey, Allele-selective inhibition of huntingtin expression by switching to an miRNA-like RNAi mechanism, Chem. Biol. 17 (2010) 1183–1188, https://doi.org/10.1016/j.chembiol.2010.10.013.
[89] A. Fiszer, A. Mykowska, W.J. Krzyzosiak, Inhibition of mutant huntingtin expression by RNA duplex targeting expanded CAG repeats, Nucleic Acids Res. 39 (2011) 5578–5585, https://doi.org/10.1093/nar/gkr156.
[90] J. Hu, J. Liu, D. Yu, Y. Chu, D.R. Corey, Mechanism of allele-selective inhibition of huntingtin expression by duplex RNAs that target CAG repeats: function through the RNAi pathway, Nucleic Acids Res. 40 (2012) 11270–11280, https://doi.org/10.1093/nar/gks907.
[91] A. Fiszer, M. Olejniczak, P. Galka-Marciniak, A. Mykowska, W.J. Krzyzosiak, Self-duplexing CUG repeats selectively inhibit mutant huntingtin expression, Nucleic Acids Res. 41 (2013) 10426–10437, https://doi.org/10.1093/nar/gkt825.
[92] J. Liu, H. Pendergraff, K.J. Narayanannair, J.G. Lackey, S. Kuchimanchi, K. G. Rajeev, M. Manoharan, J. Hu, D.R. Corey, RNA duplexes with abasic substitutions are potent and allele-selective inhibitors of huntingtin and ataxin-3 expression, Nucleic Acids Res. 41 (2013) 8788–8801, https://doi.org/10.1093/nar/gkt594.
[93] D. Yu, H. Pendergraff, J. Liu, H.B. Kordasiewicz, D.W. Cleveland, E.E. Swayze, W. F. Lima, S.T. Crooke, T.P. Prakash, D.R. Corey, Single-stranded RNAs use RNAi to potently and allele-selectively inhibit mutant huntingtin expression, Cell 150 (2012) 895–908, https://doi.org/10.1016/j.cell.2012.08.002.
[94] M.S. Lombardi, L. Jaspers, C. Spronkmans, C. Gellera, F. Taroni, E. Di Maria, S. D. Donato, W.F. Kaemmerer, A majority of Huntington's disease patients may be treatable by individualized allele-specific RNA interference, Exp. Neurol. 217 (2009) 312–319, https://doi.org/10.1016/j.expneurol.2009.03.004.
[95] E.L. Pfister, L. Kennington, J. Straubhaar, S. Wagh, W. Liu, M. DiFiglia, B. Landwehrmeyer, J.P. Vonsattel, P.D. Zamore, N. Aronin, Five siRNAs targeting three SNPs may provide therapy for three-quarters of Huntington's disease patients, Curr. Biol. 19 (2009) 774–778, https://doi.org/10.1016/j.cub.2009.03.030.
[96] R.L. Boudreau, I. Martins, B.L. Davidson, Artificial microRNAs as siRNA shuttles: improved safety as compared to shRNAs in vitro and in vivo, Mol. Ther. 17 (2009) 169–175, https://doi.org/10.1038/mt.2008.231.
[97] S. Aguiar, B. van der Gaag, F.A.B. Cortese, RNAi mechanisms in Huntington's disease therapy: siRNA versus shRNA, Transl. Neurodegener. 6 (2017) 30, https://doi.org/10.1186/s40035-017-0101-9.
[98] B.L. Davidson, H.L. Paulson, Molecular medicine for the brain: silencing of disease genes with RNA interference, Lancet Neurol. 3 (3) (2004) 145–149, https://doi.org/10.1016/S1474-4422(04)00678-7.
[99] D. Trochet, B. Prudhon, S. Vassilopoulos, M. Bitoun, Therapy for dominant inherited diseases by allele-specific RNA interference: successes and pitfalls, Curr. Gene. Ther. 15 (2015) 503–510, https://doi.org/10.2174/1566523215666150812115730.
[100] V. Drouet, M. Ruiz, D. Zala, M. Feyeux, G. Auregan, K. Cambon, L. Troquier, J. Carpentier, S. Aubert, N. Merienne, F. Bourgois-Rocha, R. Hassig, M. Rey, N. Dufour, F. Saudou, A.L. Perrier, P. Hantraye, N. Déglon, Allele-specific silencing of mutant huntingtin in rodent brain and human stem cells, PloS. One 9 (2014) e99341, https://doi.org/10.1371/journal.pone.0099341.
[101] A. Kotowska-Zimmer, Y. Ostrovska, M. Olejniczak, Universal RNAi Triggers for the Specific Inhibition of Mutant Huntingtin, Atrophin-1, Ataxin-3, and Ataxin-7 Expression, Mol. Ther. Nucleic Acids 19 (2020) 562–571, https://doi.org/10.1016/j.omtn.2019.12.012.
[102] S. Meghan, A. Neil, RNA-and DNA-based therapies for Huntington's disease, in: D. C. Lo, R.E. Hughes (Eds.), Neurobiology of Huntington's Disease: Applications to Drug Discovery, CRC. Press/Taylor & Francis., Boca Raton (FL), 2011.
[103] C. Rinaldi, M.J. Wood, Antisense oligonucleotides: the next frontier for treatment of neurological disorders, Nat. Rev. Neurol. 14 (2018) 9–21, https://doi.org/10.1038/nrneurol.2017.148.
[104] K.R.Q. Lim, T. Yokota, Invention and early history of gapmers, in: Yokota, T. Maruyama, Rika. (Eds.), Gapmers: Methods and Protocols, Methods. Mol. Biol., New Jersey, 2020, pp. 3–19.
[105] A.C. Silva, D.D. Lobo, I.M. Martins, S.M. Lopes, C. Henriques, S.P. Duarte, J. C. Dodart, R.J. Nobre, L. Pereira de Almeida, Antisense oligonucleotide therapeutics in neurodegenerative diseases: the case of polyglutamine disorders, Brain 143 (2020) 407–429, https://doi.org/10.1093/brain/awz328.
[106] J.B. Carroll, S.C. Warby, A.L. Southwell, C.N. Doty, S. Greenlee, N. Skotte, G. Hung, C.F. Bennett, S.M. Freier, M.R. Hayden, Potent and selective antisense oligonucleotides targeting single-nucleotide polymorphisms in the Huntington disease gene / allele-specific silencing of mutant huntingtin, Mol. Ther. 19 (2011) 2178–2185, https://doi.org/10.1038/mt.2011.201.
[107] A.L. Southwell, H.B. Kordasiewicz, D. Langbehn, N.H. Skotte, M.P. Parsons, E. B. Villanueva, N.S. Caron, M.E. Østergaard, L.M. Anderson, Y. Xie, L.D. Cengio, H. Findlay-Black, C.N. Doty, B. Fitsimmons, E.E. Swayze, P.P. Seth, L. A. Raymond, C. Frank Bennett, M.R. Hayden, Huntingtin suppression restores cognitive function in a mouse model of Huntington's disease, Sci. Transl. Med. 10 (2018) eaar3959, https://doi.org/10.1126/scitranslmed.aar3959.
[108] K.T. Gagnon, H.M. Pendergraff, G.F. Deleavey, E.E. Swayze, P. Potier, J. Randolph, E.B. Roesch, J. Chattopadhyaya, M.J. Damha, C.F. Bennett, C. Montaillier, M. Lemaitre, D.R. Corey, Allele-selective inhibition of mutant huntingtin expression with antisense oligonucleotides targeting the expanded CAG repeat, Biochemistry 49 (2010) 10166–10178, https://doi.org/10.1021/bi101208k.
[109] M.E. Østergaard, A.L. Southwell, H. Kordasiewicz, A.T. Watt, N.H. Skotte, C. N. Doty, K. Vaid, E.B. Villanueva, E.E. Swayze, C.F. Bennett, M.R. Hayden, P. P. Seth, Rational design of antisense oligonucleotides targeting single nucleotide polymorphisms for potent and allele selective suppression of mutant Huntingtin in the CNS, Nucleic Acids Res. 41 (2013) 9634–9650, https://doi.org/10.1093/nar/gkt725.
[110] A.C. Silva, D.D. Lobo, I.M. Martins, S.M. Lopes, C. Henriques, S.P. Duarte, J. C. Dodart, R.J. Nobre, L. Pereira de Almeida, Antisense oligonucleotide therapeutics in neurodegenerative diseases: the case of polyglutamine disorders, Brain 143 (2) (2020) 407–429, https://doi.org/10.1093/brain/awz328.
[111] J.B. Carroll, S.C. Warby, A.L. Southwell, C.N. Doty, S. Greenlee, N. Skotte, G. Hung, C.F. Bennett, S.M. Freier, M.R. Hayden, Potent and selective antisense oligonucleotides targeting single-nucleotide polymorphisms in the Huntington disease gene / allele-specific silencing of mutant huntingtin, Mol. Ther. 19 (2011) 2178–2185, https://doi.org/10.1038/mt.2011.201.
[112] N.H. Skotte, A.L. Southwell, M.E. Østergaard, J.B. Carroll, S.C. Warby, C.N. Doty, E. Petoukhov, K. Vaid, H. Kordasiewicz, A.T. Watt, S.M. Freier, G. Hung, P. P. Seth, C.F. Bennett, E.E. Swayze, M.R. Hayden, Allele-specific suppression of mutant huntingtin using antisense oligonucleotides: providing a therapeutic option for all Huntington disease patients, PloS. One 9 (2014) e107434, https://doi.org/10.1371/journal.pone.0107434.
[113] N.A. Datson, A. González-Barriga, E. Kourkouta, R. Weij, J. van de Giessen, S. Mulders, O. Kontkanen, T. Heikkinen, K. Lehtimäki, J.C. van Deutekom, The expanded CAG repeat in the huntingtin gene as target for therapeutic RNA modulation throughout the HD mouse brain, PLoS. One 12 (2017) e0171127, https://doi.org/10.1371/journal.pone.0171127.
[114] A.L. Southwell, N.H. Skotte, H.B. Kordasiewicz, M.E. Østergaard, A.T. Watt, J. B. Carroll, C.N. Doty, E.B. Villanueva, E. Petoukhov, K. Vaid, Y. Xie, S.M. Freier, E.E. Swayze, P.P. Seth, C.F. Bennett, M.R. Hayden, In vivo evaluation of candidate allele-specific mutant huntingtin gene silencing antisense oligonucleotides, Mol. Ther. 22 (2014) 2093–2106, https://doi.org/10.1038/mt.2014.153.
[115] E. Doxakis, Therapeutic antisense oligonucleotides for movement disorders, Med. Res. Rev. 41 (2021) 2656–2688, https://doi.org/10.1002/med.21706.
[116] C.F. Bennett, H.B. Kordasiewicz, D.W. Cleveland, Antisense drugs make sense for neurological diseases, Annu. Rev. Pharmacol. Toxicol. 61 (2021) 831–852, https://doi.org/10.1146/annurev-pharmtox-010919-023738.
[117] X. Sun, L.O. Marque, Z. Cordner, J.L. Pruitt, M. Bhat, P.P. Li, G. Kannan, E. E. Ladenheim, T.H. Moran, R.L. Margolis, D.D. Rudnicki, Phosphorodiamidate morpholino oligomers suppress mutant huntingtin expression and attenuate neurotoxicity, Hum. Mol. Genet. 23 (2014) 6302–6317, https://doi.org/10.1093/hmg/ddu349.
[118] L.H. Wang, Z.H. Qin, Animal models of Huntington's disease: implications in uncovering pathogenic mechanisms and developing therapies, Acta Pharmacol. Sin. 27 (2006) 1287–1302, https://doi.org/10.1111/j.1745-7254.2006.00410.x.
[119] M.A. Pouladi, A.J. Morton, M.R. Hayden, Choosing an animal model for the study of Huntington's disease, Nat. Rev. Neurosci. 14 (2013) 708–721, https://doi.org/10.1038/nrn3570.
[120] J. Hu, M. Matsui, D.R. Corey, Allele-selective inhibition of mutant huntingtin by peptide nucleic acid-peptide conjugates, locked nucleic acid, and small interfering RNA, Ann. N. Y. Acad. Sci. 1175 (2009) 24–31, https://doi.org/10.1111/j.1749-6632.2009.04975.x.
[121] R.M. Friedlander, V.D. Caro, D.L. Carlisle, N.R. Monks, D.M. Stoltzfus, N. L. Bennett, Y. Suofu, J. Kim, F. Li, X. Wang, S.M. Yeh, O.V. Baranov, S.V. Baranov, B.A. Badeau, J.W. Gleaton, G.A. Cintrón-Rosado, R.R. Dyer, D.A. Stephan, W. R. Mann, S.E. Rojas-Caro, I06 Allele-selective inhibition of mutant htt transcription throughout the brain after subcutaneous administration of a patrol™-enabled investigational genetic therapy in the r6/2 Huntington's disease transgenic mouse model, J. Neurol. Neurosurg. Psychiatry (2022), https://doi.org/10.1136/jnnp-2022-ehdn.232.
[122] A.M. Quemener, M.L. Centomo, S.L. Sax, R. Panella, Small drugs, huge impact: the extraordinary impact of antisense oligonucleotides in research and drug development, Molecules 27 (2022) 536, https://doi.org/10.3390/molecules27020536.

[123] M.M. Evers, L.J. Toonen, W.M. van Roon-Mom, Antisense oligonucleotides in therapy for neurodegenerative disorders, Adv. Drug. Deliv. Rev. 87 (2015) 90–103, https://doi.org/10.1016/j.addr.2015.03.008.
[124] D.A. Braasch, D.R. Corey, Locked nucleic acid (LNA): fine-tuning the recognition of DNA and RNA, Chem. Biol. 8 (2001) 1–7, https://doi.org/10.1016/s1074-5521(00)00058-2.
[125] R. Maruyama, T. Yokota, Knocking down long noncoding RNAs using antisense oligonucleotide gapmers, in: T. Yokota, Rika Maruyama (Eds.), Gapmers: Methods and Protocols, Methods. Mol. Biol., New Jersey, 2020, pp. 49–56.
[126] M.E. Østergaard, B. Gerland, J.M. Escudier, E.E. Swayze, P.P. Seth, Differential effects on allele selective silencing of mutant huntingtin by two stereoisomers of α,β-constrained nucleic acid, ACS Chem. Biol. 9 (2014) 1975–1979, https://doi.org/10.1021/cb5003027.
[127] M.E. Østergaard, J. Nichols, T.A. Dwight, W. Lima, M.E. Jung, E.E. Swayze, P. P. Seth, Fluorinated nucleotide modifications modulate allele selectivity of SNP-targeting antisense oligonucleotides, Mol. Ther. Nucleic Acids 7 (2017) 20–30, https://doi.org/10.1016/j.omtn.2017.02.001.
[128] J. Hu, D.W. Dodd, R.H. Hudson, D.R. Corey, Cellular localization and allele-selective inhibition of mutant huntingtin protein by peptide nucleic acid oligomers containing the fluorescent nucleobase [bis-o-(aminoethoxy)phenyl] pyrrolocytosine, Med. Chem. Lett. 19 (2009) 6181–6184, https://doi.org/10.1016/j.bmcl.2009.09.004.
[129] K.T. Gagnon, J.K. Watts, H.M. Pendergraff, C. Montaillier, D. Thai, P. Potier, D. R. Corey, Antisense and antigene inhibition of gene expression by cell-permeable oligonucleotide-oligospermine conjugates, J. Am. Chem. Soc. 133 (2011) 8404–8407, https://doi.org/10.1021/ja200312y.
[130] T. Aslesh, T. Yokota, Development of antisense oligonucleotide gapmers for the treatment of Huntington's disease, in: Yokota, T. Maruyama, Rika (Eds.), Gapmers: Methods and Protocols, Methods. Mol. Biol., New Jersey, 2020, pp. 57–67.
[131] D.R. Scoles, E.V. Minikel, S.M. Pulst, Antisense oligonucleotides: a primer, Neurol. Genet. 5 (2019) e323, https://doi.org/10.1212/NXG.0000000000000323.
[132] S.M. Hersch, G. Schifitto, D. Oakes, A.L. Bredlau, C.M. Meyers, R. Nahin, H. D. Rosas, Huntington Study Group CREST-E Investigators, Coordinators, The CREST-E study of creatine for Huntington disease: a randomized controlled trial, Neurology 89 (6) (2017) 594–601, https://doi.org/10.1212/WNL.0000000000004209.
[133] C. Sheridan, Questions swirl around failures of disease-modifying Huntington's drugs, Nat. Biotechnol. 39 (2021) 650–652, https://doi.org/10.1038/s41587-021-00955-y.
[134] M.E. Rook, A.L. Southwell, Antisense oligonucleotide therapy: from design to the Huntington disease clinic, BioDrugs 36 (2022) 105–119, https://doi.org/10.1007/s40259-022-00519-9.
[135] S.J. Tabrizi, C. Estevez-Fraga, W.M. van Roon-Mom, M.D. Flower, R.I. Scahill, E. J. Wild, I. Muñoz-Sanjuan, C. Sampaio, A.E. Rosser, B.R. Leavitt, Potential disease-modifying therapies for Huntington's disease: lessons learned and future opportunities, Lancet Neurol. 21 (7) (2022) 645–658, https://doi.org/10.1016/S1474-4422(22)00121-1.
[136] K. Kingwell, Double setback for ASO trials in Huntington disease, Nat. Rev. Drug. Discov. 20 (6) (2021) 412–413, https://doi.org/10.1038/d41573-021-00088-6.
[137] K.D. Warner, C.E. Hajdin, K.M. Weeks, Principles for targeting RNA with drug-like small molecules, Nat. Rev. Drug. Discov. 17 (2018) 547–558, https://doi.org/10.1038/nrd.2018.93.
[138] E. Khan, S. Biswas, S.K. Mishra, R. Mishra, S. Samanta, A. Mishra, A. Tawani, A. Kumar, Rationally designed small molecules targeting toxic CAG repeat RNA that causes Huntington's disease (HD) and spinocerebellar ataxia (SCAs), Biochimie 163 (2019) 21–32, https://doi.org/10.1016/j.biochi.2019.05.001.
[139] E. Khan, A. Tawani, S.K. Mishra, A.K. Verma, A. Upadhyay, M. Kumar, R. Sandhir, A. Mishra, A. Kumar, Myricetin reduces toxic level of CAG repeats RNA in Huntington's disease (HD) and spino cerebellar ataxia (SCAs), Acs. Chem. Biol. 13 (2018) 180–188, https://doi.org/10.1021/acschembio.7b00699.
[140] E. Khan, S.K. Mishra, R. Mishra, A. Mishra, A. Kumar, Discovery of a potent small molecule inhibiting Huntington's disease (HD) pathogenesis via targeting CAG repeats RNA and Poly Q protein, Sci. Rep. 9 (2019) 16872, https://doi.org/10.1038/s41598-019-53410-z.
[141] K.H. Morelli, Q. Wu, M.L. Gosztyla, H. Liu, M. Yao, C. Zhang, J. Chen, R. J. Marina, K. Lee, K.L. Jones, M.Y. Huang, A. Li, C. Smith-Geater, L.M. Thompson, W. Duan, G.W. Yeo, An RNA-targeting CRISPR-Cas13d system alleviates disease-related phenotypes in Huntington's disease models, Nat. Neurosci. 26 (2023) 27–38, https://doi.org/10.1038/s41593-022-01207-1.
[142] K.H. Morelli, Q. Wu, M.L. Gosztyla, H. Liu, C. Zhang, J. Chen, R. Marina, K. Lee, K.L. Jones, W. Duan, G.W. Yeo, RNA-targeting CRISPR/Cas13d system eliminates disease-related phenotypes in pre-clinical models of huntington's disease, BioRxiv 01 (2022) 2022, https://doi.org/10.1101/2022.01.23.477417.
[143] J.R. Thomas, P.J. Hergenrother, Targeting RNA with small molecules, Chem. Rev. 108 (2008) 1171–1224, https://doi.org/10.1021/cr0681546.
[144] D.P. Bondeson, C.M. Crews, Targeted protein degradation by small molecules, Annu. Rev. Pharmacol. Toxicol. 57 (2017) 107–123, https://doi.org/10.1146/annurev-pharmtox-010715-103507.
[145] N. Mizushima, E. White, D.C. Rubinsztein, Breakthroughs and bottlenecks in autophagy research, Trends Mol. Med. 27 (2021) 835–838, https://doi.org/10.1016/j.molmed.2021.06.012.
[146] L. Wang, D.J. Klionsky, H.M. Shen, The emerging mechanisms and functions of microautophagy, Nat. Rev. Mol. Cell. Biol. 24 (2023) 186–203, https://doi.org/10.1038/s41580-022-00529-z.
[147] S. Kaushik, A.M. Cuervo, The coming of age of chaperone-mediated autophagy, Nat. Rev. Mol. Cell. Biol. 19 (2018) 365–381, https://doi.org/10.1038/s41580-018-0001-6.
[148] S. Kaushik, A.M. Cuervo, Chaperone-mediated autophagy: a unique way to enter the lysosome world, Trends Cell. Biol. 22 (2012) 407–417, https://doi.org/10.1016/j.tcb.2012.05.006.
[149] A. Williams, S. Sarkar, P. Cuddon, E.K. Ttofi, S. Saiki, F.H. Siddiqi, L. Jahreiss, A. Fleming, D. Pask, P. Goldsmith, C.J. O'Kane, R.A. Floto, D.C. Rubinsztein, Novel targets for Huntington's disease in an mTOR-independent autophagy pathway, Nat. Chem. Biol. 4 (2008) 295–305, https://doi.org/10.1038/nchembio.79.
[150] H. Jeong, F. Then, T.J. Melia, Jr, J.R. Mazzulli, L. Cui, J.N. Savas, C. Voisine, P. Paganetti, N. Tanese, A.C. Hart, A. Yamamoto, D. Krainc, Acetylation targets mutant huntingtin to autophagosomes for degradation, Cell 137 (2009) 60–72, https://doi.org/10.1016/j.cell.2009.03.018.
[151] F. Yue, W. Li, J. Zou, Q. Chen, G. Xu, H. Huang, Z. Xu, S. Zhang, P. Gallinari, F. Wang, W.L. McKeehan, L. Liu, Blocking the association of HDAC4 with MAP1S accelerates autophagy clearance of mutant Huntingtin, Aging (Albany NY) 7 (2015) 839–853, https://doi.org/10.18632/aging.100818.
[152] K. Pierzynowska, L. Gaffke, A. Hać, J. Mantej, N. Niedziałek, J. Brokowska, G. Węgrzyn, Correction of Huntington's disease phenotype by genistein-induced autophagy in the cellular model, Neuromolecular. Med. 20 (2018) 112–123, https://doi.org/10.1007/s12017-018-8482-1.
[153] Z. Li, C. Wang, Z. Wang, C. Zhu, J. Li, T. Sha, L. Ma, C. Gao, Y. Yang, Y. Sun, J. Wang, X. Sun, C. Lu, M. Difiglia, Y. Mei, C. Ding, S. Luo, Y. Dang, Y. Ding, Y. Fei, B. Lu, Allele-selective lowering of mutant HTT protein by HTT–LC3 linker compounds, Nature 575 (2019) 203–209, https://doi.org/10.1038/s41586-019-1722-1.
[154] A.J. Tao, G.E. Gadbois, S.A. Buczynski, F.M. Ferguson, Targeted protein degradation: emerging concepts and protein state-specific targeting principles, Curr. Opin. Chem. Biol. 67 (2022) 102114, https://doi.org/10.1016/j.cbpa.2021.102114.
[155] J. Long, X. Luo, D. Fang, H. Song, W. Fang, H. Shan, P. Liu, B. Lu, X.M. Yin, L. Hong, M. Li, Discovery of an autophagy inducer J3 to lower mutant huntingtin and alleviate Huntington's disease-related phenotype, Cell. Biosci. 12 (2022) 167, https://doi.org/10.1186/s13578-022-00906-3.
[156] A. Cavalli, M.L. Bolognesi, A. Minarini, M. Rosini, V. Tumiatti, M. Recanatini, C. Melchiorre, Multi-target-directed ligands to combat neurodegenerative diseases, J. Med. Chem. 51 (2008) 347–372, https://doi.org/10.1021/jm7009364.
[157] S.J. Sukoff Rizzo, J.N. Crawley, Behavioral phenotyping assays for genetic mouse models of neurodevelopmental, neurodegenerative, and psychiatric disorders, Annu. Rev. Anim. Biosci. 5 (2017) 371–389, https://doi.org/10.1146/annurev-animal-022516-022754.
[158] J. Li, J.J. Røise, M. He, R. Das, N. Murthy, Non-viral strategies for delivering genome editing enzymes, Adv. Drug. Deliv. Rev. 168 (2021) 99–117, https://doi.org/10.1016/j.addr.2020.09.004.
[159] L. Wrobel, S.M. Hill, A. Djajadikerta, M. Fernandez-Estevez, C. Karabiyik, A. Ashkenazi, V.J. Barratt, E. Stamatakou, A. Gunnarsson, T. Rasmusson, E. W. Miele, N. Beaton, R. Bruderer, Y. Feng, L. Reiter, M.P. Castaldi, R. Jarvis, K. Tan, R.W. Bürli, D.C. Rubinsztein, Compounds activating VCP D1 ATPase enhance both autophagic and proteasomal neurotoxic protein clearance, Nat. Commun. 13 (2022) 4146, https://doi.org/10.1038/s41467-022-31905-0.
[160] T.W. Miller, A. Messer, Intrabody applications in neurological disorders: progress and future prospects, Mol. Ther. 12 (2005) 394–401, https://doi.org/10.1016/j.ymthe.2005.04.003.
[161] Y. Li, C. Guha, P. Asp, X. Wang, T.L. Tchaikovskya, K. Kim, M. Mendel, G.J. Cost, D.H. Perlmutter, N. Roy-Chowdhury, I.J. Fox, A. Conway, J. Roy-Chowdhury, Resolution of hepatic fibrosis after ZFN-mediated gene editing in the PiZ mouse model of human α1-antitrypsin deficiency, Hepatol. Commun. 7 (2023) e0070, https://doi.org/10.1097/HC9.0000000000000070.
[162] P.O. Bauer, A. Goswami, H.K. Wong, M. Okuno, M. Kurosawa, M. Yamada, H. Miyazaki, G. Matsumoto, Y. Kino, Y. Nagai, N. Nukina, Harnessing chaperone-mediated autophagy for the selective degradation of mutant huntingtin protein, Nat. Biotechnol. 28 (2010) 256–263, https://doi.org/10.1038/nbt.1608.
[163] H. Song, C. Wang, C. Zhu, Z. Wang, H. Yang, P. Wu, X. Cui, J. Botas, Y. Dang, Y. Ding, Y. Fei, B. Lu, Suppression of toxicity of the mutant huntingtin protein by its interacting compound, desonide, Proc. Natl. Acad. Sci. U. S. A. 119 (2022) e2114303119, https://doi.org/10.1073/pnas.2114303119.
[164] M. Coufal, M.M. Maxwell, D.E. Russel, A.M. Amore, S.M. Altmann, Z. R. Hollingsworth, A.B. Young, D.E. Housman, A.G. Kazantsev, Discovery of a novel small-molecule targeting selective clearance of mutant huntingtin fragments, J. Biomol. Screen. 12 (2007) 351–360, https://doi.org/10.1177/1087057107299428.
[165] S. Hyun, D. Shin, Chemical-mediated targeted protein degradation in neurodegenerative diseases, Life 11 (2021) 607, https://doi.org/10.3390/life11070607.
[166] K.K. Dev, Making protein interactions druggable: targeting PDZ domains, Nat. Rev. Drug. Discov. 3 (2004) 1047–1056, https://doi.org/10.1038/nrd1578.
[167] J.M. Lecerf, T.L. Shirley, Q. Zhu, A. Kazantsev, P. Amersdorfer, D.E. Housman, A. Messer, J.S. Huston, Human single-chain Fv intrabodies counteract in situ huntingtin aggregation in cellular models of Huntington's disease, Proc. Natl. Acad. Sci. U. S. A. 98 (2001) 4764–4769, https://doi.org/10.1073/pnas.071058398.
[168] T.W. Miller, C. Zhou, S. Gines, M.E. MacDonald, N.D. Mazarakis, G.P. Bates, J. S. Huston, A. Messer, A human single-chain Fv intrabody preferentially targets

amino-terminal huntingtin fragments in striatal models of Huntington's disease, Neurobiol. Dis. 19 (2005) 47–56, https://doi.org/10.1016/j.nbd.2004.11.003.
[169] W.J. Wolfgang, T.W. Miller, J.M. Webster, J.S. Huston, L.M. Thompson, J. L. Marsh, A. Messer, Suppression of Huntington's disease pathology in Drosophila by human single-chain Fv antibodies, Proc. Natl. Acad. Sci. U. S. A. 102 (2005) 11563–11568, https://doi.org/10.1073/pnas.0505321102.
[170] D.C. Butler, A. Messer, Bifunctional anti-huntingtin proteasome-directed intrabodies mediate efficient degradation of mutant huntingtin exon 1 protein fragments, PloS. One 6 (2011) e29199, https://doi.org/10.1371/journal.pone.0029199.
[171] A.L. Southwell, A. Khoshnan, D.E. Dunn, C.W. Bugg, D.C. Lo, P.H. Patterson, Intrabodies binding the proline-rich domains of mutant huntingtin increase its turnover and reduce neurotoxicity, J. Neurosci. 28 (2008) 9013–9020, https://doi.org/10.1523/JNEUROSCI.2747-08.2008.
[172] A.L. Southwell, J. Ko, P.H. Patterson, Intrabody gene therapy ameliorates motor, cognitive, and neuropathological symptoms in multiple mouse models of Huntington's disease, J. Neurosci. 29 (2009) 13589–13602, https://doi.org/10.1523/JNEUROSCI.4286-09.2009.
[173] C.E. Wang, H. Zhou, J.R. McGuire, V. Cerullo, B. Lee, S.H. Li, X.J. Li, Suppression of neuropil aggregates and neurological symptoms by an intracellular antibody implicates the cytoplasmic toxicity of mutant huntingtin, J. Cell. Biol. 181 (2008) 803–816, https://doi.org/10.1083/jcb.200710158.
[174] A.L. Marschall, S. Dübel, Antibodies inside of a cell can change its outside: can intrabodies provide a new therapeutic paradigm? Comput. Struct. Biotechnol. J. 14 (2016) 304–308, https://doi.org/10.1016/j.csbj.2016.07.003.
[175] D. Dash, T.A. Mestre, Therapeutic update on Huntington's disease: symptomatic treatments and emerging disease-modifying therapies, Neurotherapeutics 17 (2020) 1645–1659, https://doi.org/10.1007/s13311-020-00891-w.
[176] A.J. Morton, E.A. Skillings, N.I. Wood, Z. Zheng, Antagonistic pleiotropy in mice carrying a CAG repeat expansion in the range causing Huntington's disease, Sci. Rep. 9 (2019) 37, https://doi.org/10.1038/s41598-018-37102-8.
[177] S. Ramaswamy, J.L. McBride, J.H. Kordower, Animal models of Huntington's disease, Ilar. J. 48 (2007) 356–373, https://doi.org/10.1093/ilar.48.4.356.
[178] Y. Wang, K.F. Bruggeman, S. Franks, V. Gautam, S.I. Hodgetts, A.R. Harvey, R. J. Williams, D.R. Nisbet, Is viral vector gene delivery more effective using biomaterials, Adv. Healthc. Mater. 10 (1) (2021) 2001238, https://doi.org/10.1002/adhm.202001238.
[179] J. Saraiva, R.J. Nobre, L.P. de Almeida, Gene therapy for the CNS using AAVs: the impact of systemic delivery by AAV9, J. Control. Release 241 (2016) 94–109, https://doi.org/10.1016/j.jconrel.2016.09.011.